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Department of Software Engineering
SWE 444 - Software Construction Laboratory



واجه
WAJEH

Final Report

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Abstract

WAJEH is a mobile application designed to help individuals improve their communication skills and build confidence in situations such as interviews, presentations, and public speaking. The system provides users with practical tools, including live training sessions with specialists, guided 4-7-8 breathing exercises to reduce anxiety, and social challenges that encourage gradual skill development.

The system integrates speech analysis using Azure Speech-to-Text, allowing users to record their speeches and convert them into text. The system then analyzes the text using custom algorithms and mathematical calculations to provide feedback on elements like pace, words, and speech patterns, and track users' progress over time. Through a points-based reward system and an admin dashboard for managing specialists and user reports, **WAJEH** offers a supportive and structured environment that enables users to communicate more effectively and confidently.

1. Introduction

1.1 Problem Statement

Many individuals face significant challenges in communicating effectively in various social, academic, and professional contexts. Social anxiety, fear of negative judgment, and lack of practical opportunities to practice public speaking often prevent people from expressing their ideas clearly and confidently. This can lead to missed opportunities in education, job interviews, workplace interactions, and networking situations. Despite the availability of online resources and general advice, many solutions fail to provide personalized, practical, and structured guidance that adapts to individual needs and tracks progress over time.

1.2 Project Idea

WAJEH is a mobile application designed to address these communication challenges by providing a structured, interactive, and personalized learning experience. The app combines guided exercises, live practice sessions, social challenges, and AI-powered speech analysis to help users improve their speaking skills. It offers real-time feedback on aspects such as pronunciation, clarity, confidence, and body language, while encouraging consistent practice through gamified challenges and social interaction. **WAJEH** is intended to be both a learning tool and a supportive environment, helping users gradually overcome their communication barriers in a measurable and engaging way.

1.3 Target Users and Stakeholders

1.3.1 Target Users

WAJEH primarily targets individuals who face challenges in public speaking, interviews, or social interactions. These users want to improve their communication skills and confidence in a safe, guided, and supportive environment. The app is designed to cater to different user needs, including beginners who want basic practice, as well as advanced users seeking professional guidance and real-time feedback.

1.3.2 Actors

- **Trainees:** Users who practice speeches, join sessions, complete exercises, and track their progress.
- **Admin:** Responsible for managing specialist accounts, handling reports, monitoring system activities.
- **Specialists:** Experts who provide live audio/video sessions and give personalized feedback to Trainees.

1.3.3 Stakeholders

Our primary stakeholder is our instructor, Ms. Nora Alzamel, Since the development of WAJEH is undertaken as a part of the Software Construction Laboratory course at the university.

1.4 Value and Market Contribution

WAJEH offers a convenient, comprehensive platform that integrates professional guidance and feedback to improve communication skills. It contributes to the market by addressing a widespread need, boosting users' confidence, and helping them perform better in real-life situations, ultimately providing measurable improvement in communication abilities.

2. Technical Details

2.1 Platform: Android

WAJEH was primarily designed for the Android platform, chosen because of its large user base and accessibility across a wide range of devices. Android's flexibility and open-source nature also made it easier to integrate with third-party services and libraries that supported features like video calls, notifications, and scheduling.

2.2 Development Platform / Framework: Flutter

We selected Flutter as our development framework because it enabled cross-platform development with a single codebase. While our initial focus was on Android, Flutter gave us the option to expand to iOS in the future without rebuilding the app from scratch. Flutter's customizable UI widgets were valuable for creating a clear and engaging interface, which was important for users dealing with challenges such as stress, speech difficulties, and stuttering. Additionally, Flutter's near-native performance ensured smooth user interactions and supported features such as progress tracking and interactive exercises.

2.3 Language: Dart

For WAJEH, we selected Dart as the programming language because it was the core language of Flutter. Dart's combination of ease of learning, stability, and high performance made it well-suited for building a reliable and responsive mobile application. Since WAJEH required smooth handling of features like scheduling calls, managing profiles, tracking progress, and processing speech input, Dart's performance and flexibility ensured these features ran efficiently.

2.4 Database: Firebase (Cloud Firestore)

Firebase was selected as the database for WAJEH because of its reliable features and seamless integration with Flutter. As a comprehensive app development platform by Google, Firebase provided all the backend services required for our app, including cloud storage. For database management, we chose Cloud Firestore because of its ease of use in storing and managing application data.

2.5 Development Environment: Visual Studio Code

We chose Visual Studio Code to develop WAJEH because of its lightweight yet powerful environment, which made it highly suitable for Flutter development. With its rich ecosystem of extensions, including Flutter and Dart plugins, VS Code streamlined our workflow by providing syntax highlighting, intelligent code completion, and robust debugging tools. These features were particularly valuable as we implemented complex functionalities such as scheduling calls, managing user profiles, and integrating Firebase services.

2.6 Development Environment: Android Studio

We used Android Studio mainly for testing and running WAJEH on Android emulators. Its built-in device simulator allowed us to preview how the app would perform on different Android devices without needing physical hardware. While Visual Studio Code was our primary editor, Android Studio provided valuable tools for performance monitoring, debugging, and ensuring compatibility with various Android versions, making it a supportive companion tool in the development process.

2.7 Payment API: Stripe

For WAJEH, we selected Stripe API as the payment solution to manage transactions between specialists and trainees. Stripe was one of the most widely used and reliable payment platforms, offering secure and seamless integration with mobile applications, including Flutter. It supported multiple payment methods, including Visa, Mastercard, and Google Pay, making it flexible and user-friendly.

We chose Stripe because of its developer-friendly documentation, straightforward integration with Flutter, and strong security features that ensured safe handling of sensitive financial data. Stripe enabled trainees to pay for booked sessions directly through the app, while allowing specialists to receive their payments in a transparent and efficient manner. It also provided a refund mechanism, which allowed users to cancel and reverse payments. This simplified the payment process, built trust between both parties, and made it the ideal choice for WAJEH's payment system.[see appendix 1]

2.8 Video Call API: Agora

We chose Agora API to enable video and audio calls between specialists and trainees in WAJEH. Agora ensured reliable and low-latency communication, which was essential for interactive sessions. It also gave users the flexibility to choose between audio-only or full video calls, depending on their needs. With its seamless integration into Flutter, Agora allowed us to deliver smooth and effective real-time communication directly within the app.[see appendix 2]

2.9 Speech-to-Text API: Azure Speech-to-Text

For WAJEH, we selected the Azure Speech-to-Text API because it provided strong support for Arabic and its dialects, which was essential for our application's target users. The API could accurately convert spoken input into text in Modern Standard Arabic as well as dialects, making it highly suitable for handling speech data within the system. Moreover, Azure Speech-to-Text was easy to integrate, ensuring smooth adoption into our Flutter-based system.[see appendix 3]

2.10 Version Control: GitHub

We selected GitHub to manage our project's source code because it provided a reliable version control system. Using GitHub allowed our team to collaborate efficiently, track code changes, and ensure that everyone was working on the latest version of the project. Its integration with development tools also made it easier to manage updates and maintain the project over time.[see appendix 4]

2.11 Project Management: Trello

Trello was used for task and project management, allowing the team to organize development stages, track progress, assign responsibilities, and ensure timely completion of project milestones.[see appendix 5]

3. System Analysis

3.1 System architecture

Model-View-ViewModel (MVVM) architecture was the architecture that we chose for WAJEH. It was a modern and widely used pattern in real-world mobile applications, also suitable for Android apps, as it simplified managing complex interfaces while keeping business logic separate, making development faster, more organized, and easier to maintain.

In WAJEH's MVVM architecture, the **Model** was the component that managed the app's core data and all interactions with external services. It was responsible for handling trainee and specialist profiles, booked sessions, practice texts, speech recordings, and progress tracking.

The Model also connected the system with external services and databases. It communicated with Firebase to read and write real-time data, ensuring that information like progress or session updates was instantly synchronized across devices. It also sent and received data from external APIs: Stripe for payment processing, Agora for real-time communication, and Azure Speech-to-Text for analyzing speech recordings.

By concentrating these responsibilities inside the Model, WAJEH kept the user interface and the business logic separate. This made the View and ViewModel simpler, as they only displayed information and handled user interaction, while the Model ensured that all data and integrations remained consistent, secure, and up to date.

The **View** represented the user interface and captured user interactions. Built with Flutter, the View enabled each user type—Trainee, Specialist, and Admin—to seamlessly interact with the system. The diagram illustrated bidirectional communication between these users and the backend services, highlighting the continuous two-way flow of information that connected all parts of the system.

The **ViewModel** came between the Model and the View. It processed user input, updated the Model, and prepared data for the View. For instance, when a trainee booked a session, the ViewModel updated schedules and refreshed the interface immediately. It also handled feedback from speech analysis, reward scores, and interactions with specialists, ensuring a seamless and consistent experience.

Using the MVVM architecture allowed WAJEH to separate the user interface from the business logic, so that updates to the interface, like adding new screens or features, could be done without affecting the core functions of the app. This structure made it easier to test

each part of the app individually and to reuse components across different sections, saving time during development.

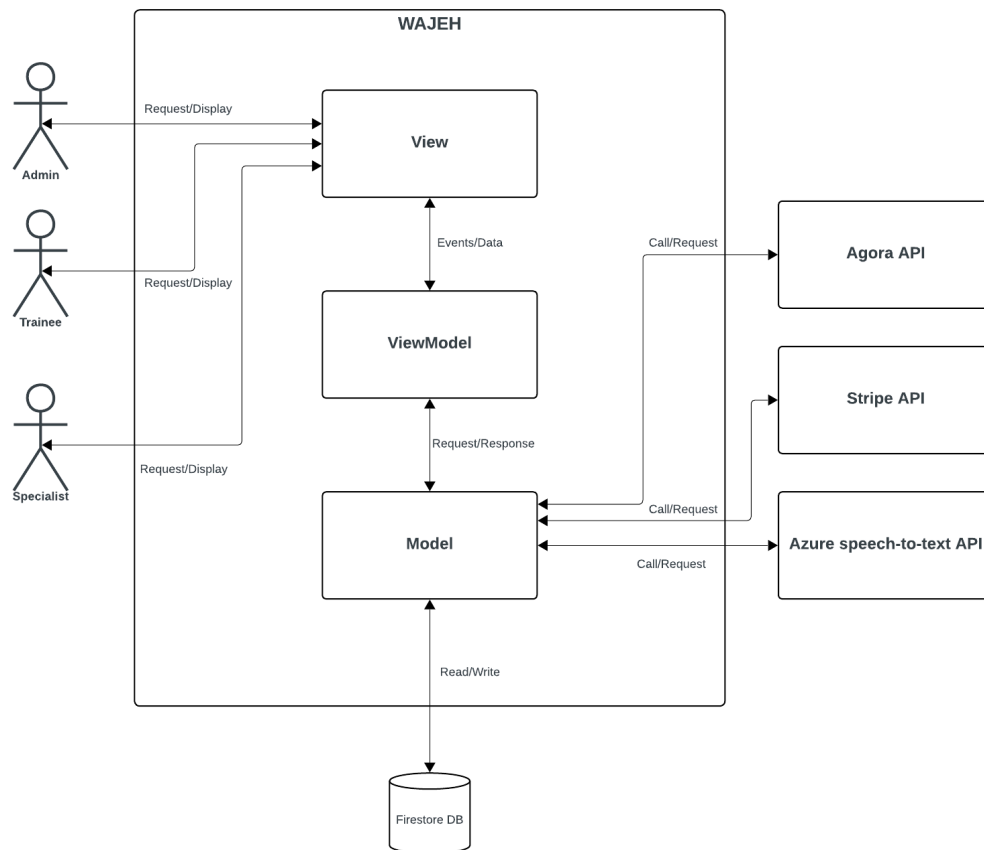


Figure [1]: WajeH system Architecture diagram

3.2 System database

Firebase Database is a real-time NoSQL that we choose to support **WAJEH's** interactive and synchronized learning experience. Firebase's capabilities align directly with our app's core functionalities, such as instant feedback, session updates, and progress tracking.

Key Firebase features that supports this choice:

1. **Real-time synchronization:** Firebase ensures low-latency, responsive updates for live sessions and user interactions that will deliver a seamless and collaborative practice environment for our trainees.
2. **Agile backend infrastructure:** By offering built-in services like reset passwords and Cloud Firestore, it enables rapid development and iteration without managing servers manually.
3. **Security and privacy:** Firebase ensures strong protection of user data through secure authentication and access control mechanisms. Importantly, **WAJEH** does not store user passwords inside the Firestore database; all authentication is handled securely through Firebase Authentication, ensuring that sensitive credentials remain encrypted and inaccessible.

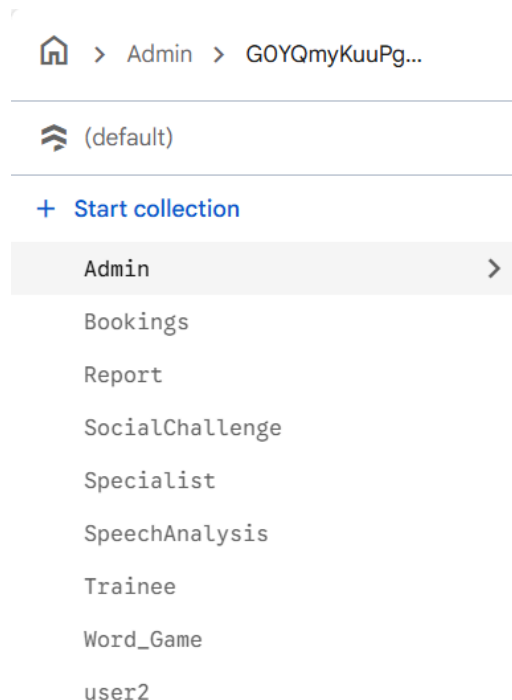


Figure [2]: WAJEH system database

We have 9 collections for storing data.

1. Admin collection

This collection stores information about system admin responsible for managing specialists, reviewing reports, and monitoring platform activities. The admin document contains the admin's email, full name (represented through firstName and lastName), and the fcmToken used for sending push notifications to their device.

</

Figure [3]: Admin collection

2. user2 collection

The user collection serves as a unified identity layer for all users in WAJEH and plays a crucial role during the sign-in process. Each document in this collection stores the user's email along with a role field that clearly distinguishes whether the user is an Admin, Specialist, or Trainee.

>

user2

>

OPLJCxcXMBc2...

More in Google Cloud

(default)

user2

OPLJCxcXMBc27Nfx5iYXkTquVjf2

+ Start collection

Admin

Bookings

Report

SocialChallenge

Specialist

SpeechAnalysis

Trainee

Word_Game

user2

+ Add document

0PLJCxcXMBc27Nfx5iYXkTquVj... >

0umwC9JimZTRUUbWh3FX4FuQs3...

3xrh5LLc58VV7aNHWRuzoBJPMO...

4hp321wI7BPYuHW5p0kuFr1xTh...

4rp5ZfLhBiXhs4K0CpKy7MjTLK...

68LXWzEyjtMpHoDghP0SuFqk03...

6E3w7ALsEbYvdwVZuwcnnna2Lc...

7koWHBy8zaM9UabVrGoVgQkeXt...

DD3Y3vSW24YCM8F0EokMrUJZoh...

G0YQmyKuuPgORx0tYpEOgtzjpe...

GwMWLxsF3JOsfuYEO2VWUAU7nC...

H3uZJhyiA3VSsrBcHD2RNxhSbM...

IZ6320acv4UBHFOUZcLQnWNqYj...

J1BT7nZwhaMdVvi2WnCcKcyM6h9...

+ Start collection

+ Add field

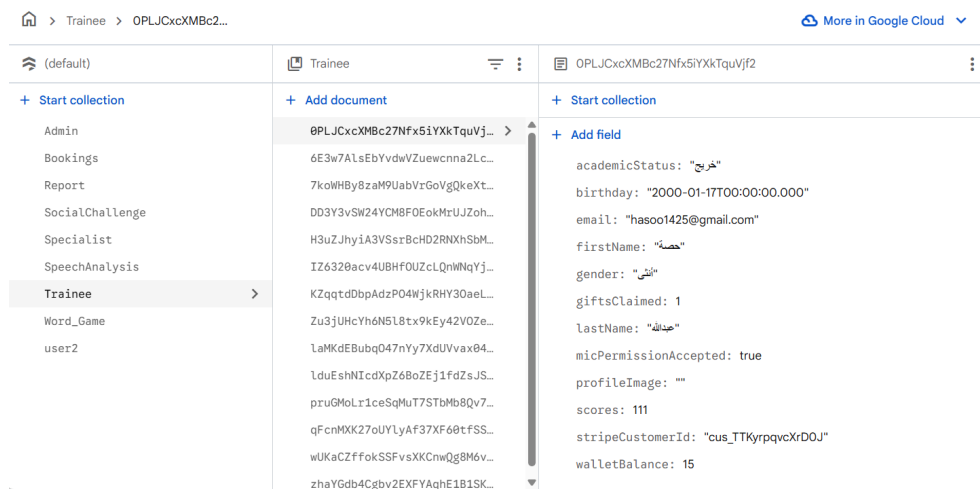
email: "hasoo1425@gmail.com"

role: 1

Figure [4]: user2 collection

3.Trainee collection

The Trainee collection includes all registered users who practice speaking, join live sessions, and track their communication progress. Each trainee document stores information that include firstName,lastName,email,academicStatus,birthday,gender,lastLevelPopup,micPermissionAccepted,profileImage,scores, and stripeCustomerd.

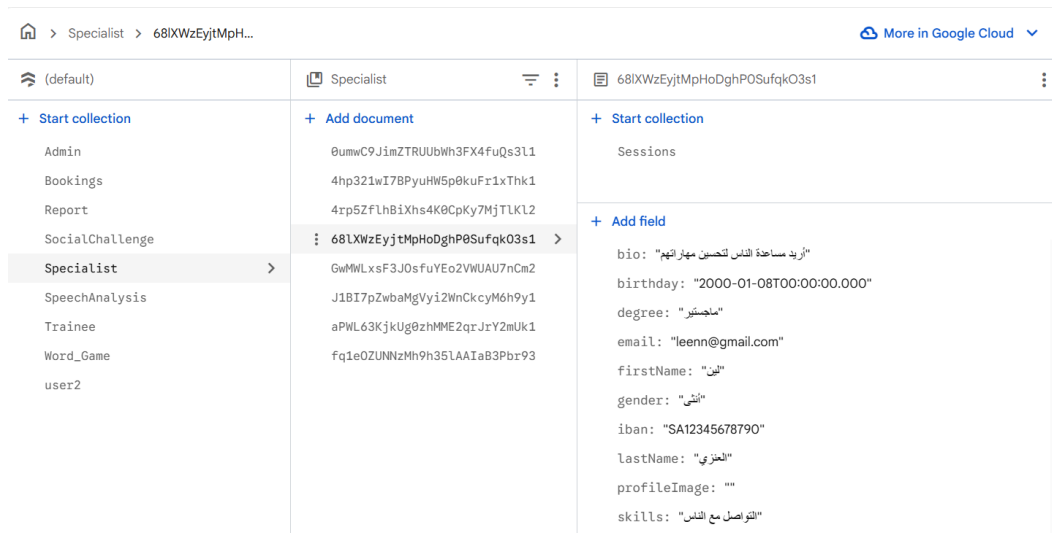


(default)	Trainee	OPLJCxcXMBc27Nfx5iYXkTquVjf2
+ Start collection	+ Add document	+ Start collection
Admin	0PLJCxcXMBc27Nfx5iYXkTquVj...	+ Add field
Bookings	6E3w7AlSbEbYvdwVZuwcna2Lc...	academicStatus: "خريج"
Report	7koWHBy8zaM9UabVrGoVgQkeXt...	birthday: "2000-01-17T00:00:00.000"
SocialChallenge	DD3Y3vSW24YCM8F0EokMrUJZoh...	email: "hasoot1425@gmail.com"
Specialist	H3uZJhyia3VSsrBcHD2RNXhSbM...	firstName: "حصه"
SpeechAnalysis	IZ6320acv4U8HFOUZcLQnWnqYj...	gender: "انثى"
Trainee	KZqqtDdbAdzP04WjkRHY30ael...	giftsClaimed: 1
Word_Game	Zu3jUHcYh6N5l8tx9kEy42VOZe...	lastName: "عبدالله"
user2	1aMKdEBubq047nYy7XdUVvax04...	micPermissionAccepted: true
	1duEshNtcdXpZ6BoZEj1fdZsJS...	profileImage: ""
	pruGMoLr1ceSqMuT7STbMb8Qv7...	scores: 111
	qFcmMXK27oUYLYAf37XF60tfSS...	stripeCustomerId: "cus_TTKyrpqvcXrD0J"
	wUKaCZfFokSSFvsXKCnwQg8M6v...	walletBalance: 15
	zhaYGdb4Cgbv2EXFYAqhE1B1SK...	

Figure [5]: Trainee collection

4.Specialist collection

This collection contains records of certified specialists who provide professional speaking sessions and personalized feedback. Each specialist document includes essential information such as the specialist's firstname, lastName, email, bio, birthday, degree ,gender, iban,profileImage,skills and status.



(default)	Specialist	68lXWzEyjtMpHoDghP0SufqkO3s1
+ Start collection	+ Add document	+ Start collection
Admin	0umwC9JimZTRUUbWh3FX4fuQs3L1	Sessions
Bookings	4hp321wI7BPYuHw5p0kuFr1xThk1	
Report	4rp5ZFlhBiXhs4K0CpKy7MjTLK12	+ Add field
SocialChallenge	68lXWzEyjtMpHoDghP0SufqkO3s1	bio: "أريد مساعدة الناس لتحسين مهاراتهم"
Specialist	GwMWLxsF3J0sfuYeo2VWUUAU7nCm2	birthday: "2000-01-08T00:00:00.000"
SpeechAnalysis	J1BI7pZwbaMgVyi2WnCcKyM6h9y1	degree: "ماجستير"
Trainee	aPWL63KjkUg0zhMME2qrJrY2mUk1	email: "leenn@gmail.com"
Word_Game	fq1e0ZUNNzhMh9h35LAAIA83Pbr93	firstName: "لين"
user2		gender: "انثى"
		iban: "SA1234567890"
		lastName: "العنزي"
		profileImage: ""
		skills: "التواصل مع الناس"

Figure [6]: Specialist collection

5.Bookings collection

The Bookings collection manages all scheduled sessions between trainees and specialists. Each booking document stores the traineeID and specialistID to link both parties,in addition to the unique SessionID, Date, CreatedAt, CallEndedAt,Duration, Price, Status, stripeCustomerId, walletTransactionId.

🏠 > Bookings > OMtmfORV3cff...

☁ More in Google Cloud

(default)

Bookings

OMtmfORV3cffZH8eBZ

+ Start collection

Admin

Bookings

Report

SocialChallenge

Specialist

SpeechAnalysis

Trainee

Word_Game

user2

+ Add document

0MtmfORV3cffZH8eBZ

2e7yRIImfvTBot9k75AG

2uPm6t8tDizsEFo9YEis

5k2GnYTiZWdzhjvxIpxL

6tuAFZbGLuXS2KqIyBUi

8m9cff9wihlFAi77vu4H

9mLYG3JcGkLNIpotnwYI

ArVdCUOKVSZwEa1S5TLu

B2Q6dSXwLyldhWSJZU4l

CmTrQ4BSnKyvfwim9fnP

GthCiFYbiebI65PBYDvf

JVMAXD6za7JaMxUtkMbN

K0oDrD5q5b6jzKcT900e

LVze4601LFBGkJ69J3UL

+ Start collection

+ Add field

CallEndedAt: November 15, 2025 at 1:04:11AM UTC+3

CallStatus: "ended"

CreatedAt: November 15, 2025 at 12:25:56AM UTC+3

Date: November 15, 2025 at 12:37:00AM UTC+3

Duration: 33

Price: 30

SessionID: "OMtmfORV3cffZH8eBZ"

SpecialistID: "68lXWzEyjtMpHoDghPOsufqkO3s1"

Status: "Paid (Wallet)"

TraineeID: "H3uZJhyIA3VSsrBcHD2RNxhSbMM2"

stripeCustomerId: "cus_TNfXZChUcSo1qL"

walletTransactionId: "cbtxn_1STUVDDBQyyKGilUjO7lr9HuY"

Figure [7]: Bookings collection

6.SocialChallenge collection

This collection contains all social challenges available in the app. Each document includes a challenge title and description, This collection supports the gamified learning experience provided by **WAJEH**.

🏠 > SocialChallenge > Document1

☁ More in Google Cloud

🔌 (default)

+ Start collection

Admin

Bookings

Report

SocialChallenge

Specialist

SpeechAnalysis

📁 SocialChallenge

+ Add document

Document1

Document10

Document2

Document3

Document4

Document5

📄 Document1

+ Start collection

+ Add field

Title: "ابداً محادثة مع شخص غريب"

description: "تحدث مع شخص لا تعرفه اليوم، سواء في العمل أو الحي أو المقهى، لتكسر حاجز "الخجل وتزيد ثقته"

Figure [8]: SocialChallenge collection

7.SpeechAnalysis collection

The SpeechAnalysis collection stores the results generated from trainees' speech recordings. After converting the audio to text using Azure Speech-to-Text, the system performs additional custom analysis and saves the output in this collection. Each document contains fields that include `userId` `manualText`, `timestamp`, `transcribedText` and `feedback`.

🏠 > SpeechAnalysis > M7wR9v14J5de... More in Google Cloud		
(default)	SpeechAnalysis	M7wR9v14J5den5OPldY
+ Start collection	+ Add document	+ Start collection
Admin	BxYUGUH6vLaLPWIjwpTp	+ Add field
Bookings	COao9W3VCA5dQPw8X7kz	feedback: "كنت تتحدث ببطء، حاول أن تتحدث بسرعة أقرب للمعدل الطبيعي. تطابق كلامك مع: "النص ممتاز جدًا"
Report	Lkjd4Epn3A0wuApi5jgd	manualText: "السلام عليكم ورحمة الله وبركاته مساء الخير كيكن شو اخباركن ان شاء الله بخير"
SocialChallenge	M7wR9v14J5den50PlcdY	timestamp: November 25, 2025 at 8:42:37 PM UTC+3
Specialist	MZh3RFmdfBiNjvylZS4m	transcribedText: "السلام عليكم ورحمة الله وبركاته. مساء الخير، كيكن؟ شو أخباركم؟ إن شاء الله بخير؟"
SpeechAnalysis	SZyl8rdjHvXrcx7NZfcS	userId: "H3uZJhyiA3VSsrBcHD2RNxhSbMM2"
Trainee	SmijjvQb08aQzNXkUjbn	

Figure [9]: SpeechAnalysis collection

8.Report collection

This Report collection organizes all reports submitted by trainees or specialists after completing a session when they encounter any kind of communication skills. Each report document includes a `bookingId`, `reportedAt`, `reportedOn`, `reporter` and `description`.

🏠 > Report > IOiYwWdk56aaf... More in Google Cloud		
(default)	Report	IOiYwWdk56aafvp6ZQg
+ Start collection	+ Add document	+ Start collection
Admin	LN/LLY3FFGF0HLYE13	+ Add field
Bookings	bvaqKhodyNxtus6ndVSN	bookingId: "AFOytF3hKHtivWofHuj"
Report	ccXkQL2nmSfW0H1XGwhn	description: "المكتربة قامت بالتعدي على بالفاظ"
SocialChallenge	dLVZewH1IWZrErZ6Fe41	reportedAt: November 9, 2025 at 2:07:51 AM UTC+3
Specialist	fQTgedBpT30bhk7vVpU2	reportedOn: "IZ6320acv4UBHfOUZcLQnWNqYj32"
SpeechAnalysis	ft5ucVGMNUJi6CwlpT2L	reporter: "68lXWzEyjtMpHoDghPOsufqkO3s1"
Trainee	iJ3BjTJuqEugMupuGHQZ	

Figure [10]: Report collection

9.Word_Game collection

This collection stores data related to the in-app speaking game designed to help trainees practice vocabulary and spontaneous speech. Each document includes a hint, level,scrambled and unscrambled

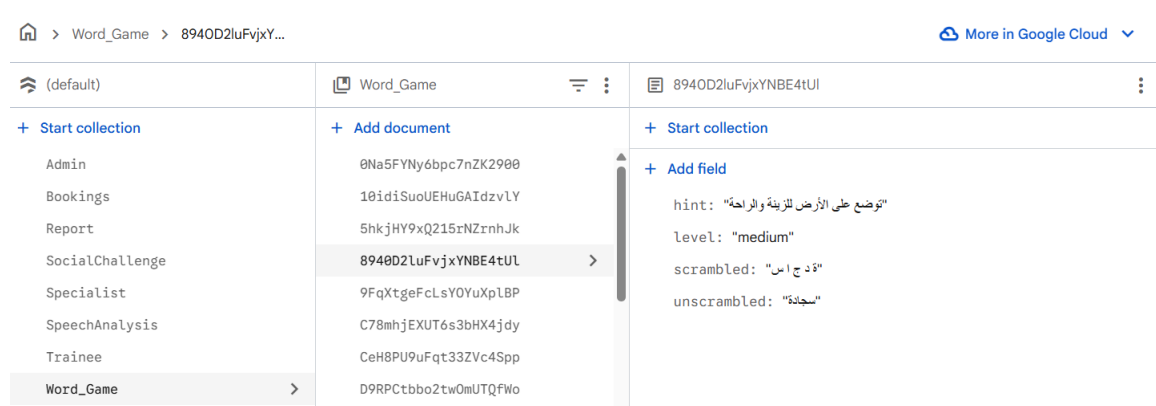


Figure [11]: Word_Game collection

4. Final Product Backlog

In this section, the team should describe the status of the features in the original product backlog and the newly added features. You can refer to the example below for better understanding. It is necessary to report every change in product backlog and the status of every feature.

Definitions

- **User** : Trainee, Specialist, and Admin.
- **(4-7-8)breathing exercises** :The 4-7-8 breathing technique involves breathing in for 4 seconds, holding the breath for 7 seconds, and exhaling for 8 seconds.

Original product backlog status

Feature ID	Brief Description	Status	Comment
User features			
1	User can sign in using their email and password.	Delivered	
2	User can sign out.	Delivered	
3	User can reset their password via a link sent to registered email.	Delivered	
Trainee features			
4	Trainee can sign up using (email, first name, last name, gender, birthdate, profile picture(optional), password ,and academic status).	Delivered	
5	Trainee can view their profile details (email, first name, last name, gender, birthdate,wallet balance, profile picture, academic status).	Delivered	

6	Trainee can edit their profile details (email ,first name, last name, profile picture, academic status).	Delivered	
7	Trainee can Start guided (4-7-8)breathing exercises .	Delivered	
8	Trainee can vague in social challenges.	Delivered	
9	Trainee can view their progress(scores, skill status, rewards, Level of development,Today's summary).	Delivered	
10	Trainee can view a list of registered specialists.	Delivered	
11	Trainee can filter specialists based on their ratings, and degree.	Delivered	
12	Trainee can view a specialist's profile details (first name, last name, gender, profile picture, bio, skills, degree, average rating).	Delivered	
13	Trainee can view a specialist's available sessions with its price .	Delivered	
14	Trainee can book an available session with a specialist.	Delivered	

15	Trainee can pay via Visa, Mastercard, or Google pay, to book a session with a specialist.	Delivered	we have decided to cancel paying using Google Pay because it may cause integration issues and delays
16	Trainee can view their booked sessions.	Delivered	
17	Trainee can cancel booked sessions until 2 hours before it starts.	Delivered	
18	Trainee can reschedule sessions until 2 hours before the session starts.	Delivered	
19	Trainee can join direct video/audio calls with a specialist.	Delivered	
20	Trainee can report a Specialist after ending the session.	Delivered	
21	Trainee can add a custom text for speech practice.	Delivered	
22	Trainee can record their speech for 2 hours at most, for speech analysis.	Delivered	
23	Trainee can view detailed feedback after speech analysis.	Delivered	
24	Trainee can save speech texts with its feedback.	Delivered	

25	Trainee can view the history of past speech texts with its feedback.	Delivered	
26	Trainee can view the notification list .	Not Delivered	The team realized the feature is not useful to the user ,Because the user is receives notifications directly and there is no need to display a list of them.
27	trainee can receive a balance in the app wallet as a reward every time they collect 100 scores .	Delivered	
Specialist features			
28	Specialist can sign up using(email, password, first name, last name, gender, birthdate, profile picture(optional), bio, skills, and degree,bank iban).	Delivered	
29	Specialist can view their profile details (email, first name, last name, gender, birthdate, profile picture, bio, skills, degree, and average rating,bank account details).	Delivered	
30	Specialist can edit their profile details (email, first name, last name, profile picture, bio, skills, degree).	Delivered	
31	Specialist can publish available time slots for sessions with its price.	Delivered	
32	Specialist can edit their published time slots for sessions with their prices.	Delivered	

33	Specialist can view booked sessions.	Delivered	
34	Specialist can view booked trainees' information (name, gender, profile picture, academic status).	Delivered	
35	Specialist can cancel booked sessions until 2 hours before the session starts.	Delivered	
36	Specialist can start direct video/audio calls with a trainee.	Delivered	
37	Specialist can report a Trainee after ending the session.	Delivered	
38	Specialist can view the notification list.	Delivered	
Admin features			
39	Admin can view their profile details (first name, last name, email).	Delivered	
40	Admin can edit profile details (first name, last name, email).	Delivered	
41	Admin can view specialist's account requests that contain specialist details(email, first name, last name, gender, birthdate,	Delivered	

	profile picture, bio, skills, degree)		
42	Admin can approve specialist accounts.	Delivered	
43	Admin can disapprove specialist accounts.	Delivered	
44	Admin can view specialist's/ trainee's reports list.	Delivered	
45	Admin can suspend a specialist's/ trainee's account if they have been reported 3 times or more.	Delivered	
46	Admin can view system activities (number of trainees, number of specialists, number of booked sessions, and average Specialists rating).	Delivered	
System features			
47	System can analyze trainee's texts, to provide detailed feedback on their speech practice.	Delivered	
48	System can send notification to the Admin when a trainee/specialist is reported.	Delivered	
49		Delivered	

	System can send notification to the trainee/specialist 2 hours before the session.		
50	System can increase the trainee's scores each time they complete an activity including (breathing practice, speech practice and participating in social challenges).	Delivered	
Knowledge features			
51	Learn how to use the Flutter framework.	Delivered	
52	Learn Dart programming language.	Delivered	
53	Learn how to use Firebase.	Delivered	
54	Learn how to connect Firebase to Flutter.	Delivered	
55	Learn how to use Android studio.	Delivered	
56	Learn how to use Visual Studio Code.	Delivered	
57	Learn GitHub commands.	Delivered	
58	Learn how to connect Cal.com API to Flutter.	Not Delivered	The team determined that the feature is no longer necessary, with agreement from the team stakeholder. Note: we implemented the booking feature from scratch instead of using Cal.com as we

			initially planned for some reasons: • Building it ourselves gave us more control and flexibility over the booking logic and user experience. • Smoother integration with our existing database and authentication system.
59	Learn how to connect Stripe API to Flutter.	Delivered	
60	Learn how to connect Agora API to Flutter.	Delivered	
61	Learn how to connect Azure Speech-to-Text API to Flutter.	Delivered	
Technical features			
62	Install Flutter SDK.	Delivered	
63	Set up Firebase.	Delivered	
64	Set up an emulator.	Delivered	
65	Install Android Studio.	Delivered	
66	Configure GitHub repository.	Delivered	
67	Cal.com API integration.	Not Delivered	Not delivered for the same reason was mentioned in feature 57 comments.
68	Install Visual Studio Code.	Delivered	
69	install Dart extension in VSCode.	Delivered	
70	Install Flutter extension in VSCode.	Delivered	

71	install Firebase command-line interface (CLI).	Delivered	
72	Connect WAJEH to the firebase.	Delivered	
73	Stripe API integration.	Delivered	
74	Agora API integration.	Delivered	
75	Azure Speech-to-Text API integration.	Delivered	

New features added to the product backlog

Feature ID	Brief Description	Status	Comment
Trainee features			
1	Trainee can leave the video/audio calls with a specialist.	Delivered	The team realized this feature is needed for feature 19.
2	Trainee can view feedback provided by the specialist after the session.	Delivered	The team realized the feature is preferred to the user
3	Trainee can enable or disable their audio/video during the session	Delivered	The team realized this feature is needed for feature 19.
4	Trainee can stop the recording of their speech.	Delivered	The team realized this feature is needed to the user and complete the feature 22.
5	Trainee can delete their recording of the speech.	Delivered	The team realized the feature is more useful and preferred to the user.

6	Trainee can play a word-unscramble mini-game by rearranging letters to fix scrambled words.	Delivered	This feature may be preferred to the user.
7	Trainee can receive a wallet balance when cancelling a session.	Delivered	The team realized the feature is more useful and preferred to the user.
8	Trainee can pay for a session using their in-app wallet balance.	Delivered	The team realized the feature is more useful and preferred to the user.
9	Trainee can listen to their recorded speech.	Delivered	The team realized the feature is more useful and preferred to the user.
10	Trainee can view a scoreboard for all trainees.	Delivered	The team realized the feature is more useful and preferred to the user.
11	Trainee can delete from speech analysis history.	Delivered	The team realized the feature is more useful and preferred to the user.
Specialist features			
12	Specialist can delete their published time slots	Delivered	The team realized the feature is needed for the user.
13	Specialist can give feedback to the trainee after the session.	Delivered	The team realized the feature is more useful and preferred to the user.
14	Specialist can enable or disable their audio/video during the session	Delivered	The team realized this feature is needed for feature 36.

15	Specialist can leave the video/audio calls with a trainee.	Delivered	The team realized this feature is needed for feature 36.
16	Specialist can view the published time slots.	Delivered	The team realized the feature is needed to the user.

5. Project Plans

System Features

In this section, fill in the table with your system features and their description under each sprint [see example below]. The description should adequately explain the feature.

Definitions

- **User** : Trainee, Specialist, and Admin.
- **(4-7-8)breathing exercises** :The 4-7-8 breathing technique involves breathing in for 4 seconds, holding the breath for 7 seconds, and exhaling for 8 seconds.

Sprint no.	Release Date	Features	Description
Sprint 1	1/10/2025	sign in	User shall be able to sign in using their email and password.
		sign out.	User shall be able to log out of their account.
		Specialist sign up	Specialist shall be able to sign up using their email, password, first name, last name, gender, birthdate, profile picture (optional), bio, skills, degree, and bank account details.
		Trainee sign up	Trainee shall be able to sign up using their email, first name, last name, gender, birthdate, profile picture (optional), password, and academic status.
		Start guided (4-7-8) breathing exercises	Trainee shall be able to start guided 4-7-8 breathing exercises

		engage in social challenges	Trainee shall be able to engage in social challenges.
		view a list of registered specialists	Trainee shall be able to view the list of registered specialists.
		view a specialist's available session	Trainee shall be able to view a specialist's available sessions along with their prices
		publish available time slots	Specialist shall be able to publish available time slots for sessions with its price
Sprint 2	22/10/2025	reset password	User shall be able to reset their password via a link sent to their registered email.
		filter specialists	Trainee shall be able to filter specialists based on their ratings and degree.
		view a specialist's profile details	Trainee shall be able to view a specialist's profile details (first name, last name, gender, profile picture, bio, skills, degree, average rating).
		book a session	Trainee shall be able to book a session with a specialist.
		pay via Visa and Mastercard	Trainee shall be able to pay for the session using Visa and Mastercard.

		view booked sessions	Trainee/Specialist shall be able to view their booked sessions.
		reschedule a session	Trainee shall be able to reschedule a session until 2 hours before the session starts.
		edit published time slots.	Specialist shall be able to edit the date and time of their published sessions.
		view the published time slots.	Specialist shall be able to view their published time slots.
		view booked trainees' information	Specialist shall be able to view booked trainees' information (name, gender, profile picture, academic status), so that I can better understand the trainee.
		view specialist account requests	Admin shall be able to view specialist account requests, including their details (email, first name, last name, gender, birthdate, profile picture, bio, skills, degree).
		approve specialist accounts,	Admin shall be able to approve specialist account requests.
		disapprove specialist accounts	Admin shall be able to disapprove specialist account requests.

		receive a notification 2 hours before the session	Specialist and Trainee shall be able to receive a notification 2 hours before the session start time
Sprint 3	9/11/2025	cancel a booked session	Trainee shall be able to cancel a booked session until 2 hours before it starts.
		join direct video/audio calls	Trainee shall be able to join direct video/audio calls with a specialist.
		rate a specialist	Trainee shall be able to rate a specialist after the session.
		report a specialist	Trainee shall be able to report a specialist after the session.
		leave the video/audio call	Trainee/Specialist shall be able to leave the video/audio call at any time during the session.
		add a custom text	Trainee shall be able to add custom practice text for recording.
		record speech	Trainee shall be able to record their speech for 2 hours at most.
		delete recorded	Trainee shall be able to delete their recorded practice.
		stop recording	Trainee shall be able to stop a recording at any time.
		view the speech analysis feedback	Trainee shall be able to view speech analysis

			feedback after completing a practice recording
		save speech texts	Trainee shall be able to save speech texts with its feedback.
		pay using app wallet	Trainee shall be able to pay for sessions using their in-app wallet.
		receive a balance in wallet when cancel a session	Trainee shall be able to receive a wallet balance refund when canceling a session.
		delete an available published time slot	Specialist shall be able to delete a published available time slot.
		cancel booked sessions	Specialist shall be able to cancel booked sessions until 2 hours before they start.
		start direct video/audio calls	Specialist shall be able to start direct video/audio calls with a trainee.
		report a Trainee	Specialist shall be able to report a trainee after the session.
		view profile details	Admin shall be able to view their profile details (first name, last name, email).
		edit profile details	Admin shall be able to edit their profile details (first name, last name, email).
		view system activities	Admin shall be able to view system activities (number of trainees,

			number of specialists, number of booked sessions, and average Specialists rating).
		view specialist's/ trainee's reports list	Admin shall be able to view specialist's/ trainee's reports list.
		suspend a specialist's/ trainee's account	Admin shall be able to suspend a specialist's or trainee's account if they have been reported 3 times or more.
Sprint 4	30/11/2025	view profile details(Specialist)	Specialist shall be able to view their profile details (email, first name, last name, gender, birthdate, profile picture, bio, skills, degree, and average rating, bank iban).
		edit profile details (Specialist)	Specialist shall be able to edit their profile details (email, first name, last name, profile picture, bio, skills and degree).
		view profile details(Trainee)	Trainee shall be able to view their profile details (email, first name, last name, gender, birthdate, wallet balance , profile picture, academic status).
		edit profile details(Trainee)	Trainee shall be able to edit their profile details (email, first name, last name, profile picture, and academic status).
		receive trainee/specialist report notifications	Admin shall be able to receive notifications when trainees or specialists are reported.
		receive a balance in app wallet as a reward	Trainee shall be able to receive a balance in their app wallet as a reward each time they collect 100 scores..

		view progress (dashboard)	Trainee shall be able to view their progress(scores, skill status, rewards, Level of development,Today's summary).
		gain scores through perform breathing exercises	Trainee shall be able to gain scores through performing guided 4-7-8 breathing exercises.
		gain scores through participate in social challenges	Trainee shall be able to gain scores through participation in social challenges.
		gain scores through practice speech analysis.	Trainee shall be able to gain scores through speech analysis by recording their speech.
		gain scores through play a word-unscramble mini-game	Trainee shall be able to gain scores through playing a word-unscramble mini-game.
		view feedback from specialist	Trainee shall be able to view feedback from specialist.
		give feedback for the trainee after the session	Specialist shall be able to give feedback to the trainee after the session
		listen to recorded speech	Trainee shall be able to listen to their recorded speech.
		play a word-unscramble mini-game	Trainee shall be able to play a word-unscramble mini-game where I fix scrambled words by rearranging their letters.
		view history of past speech texts	Trainee shall be able to view the history of past speech texts with its feedback.

		delete from speech analysis history.	Trainee can delete his past saved speech analysis texts in history.
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6.Project Burndown Charts

Cumulative charts

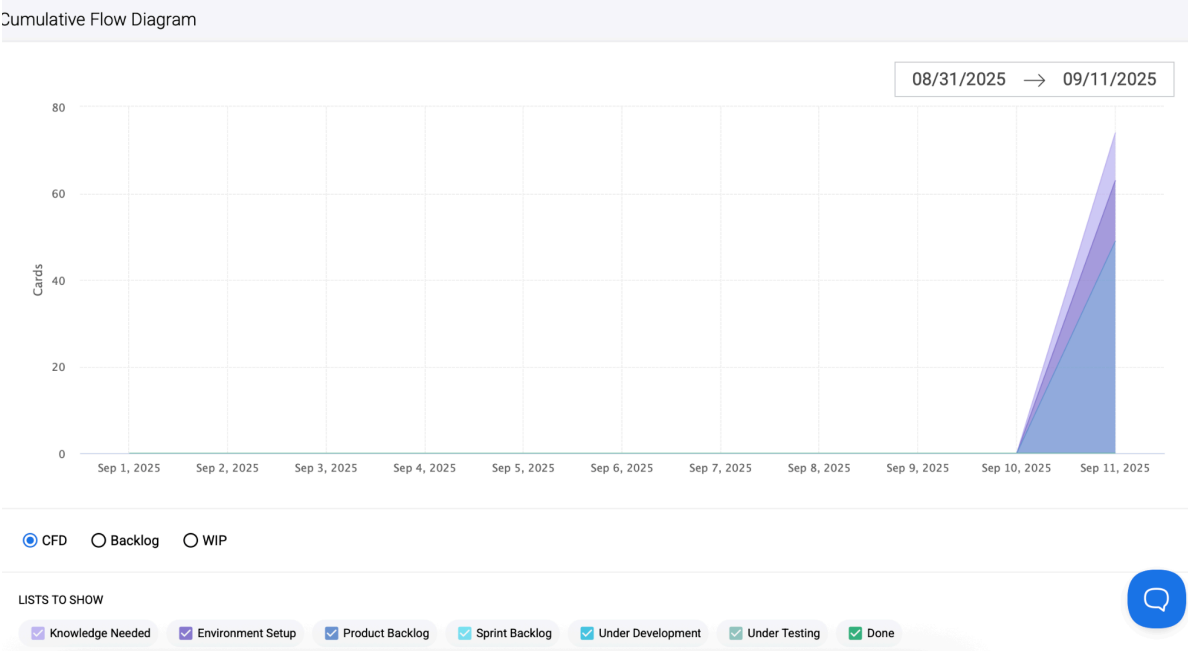


Figure [12]: Sprint 0 Cumulative Chart

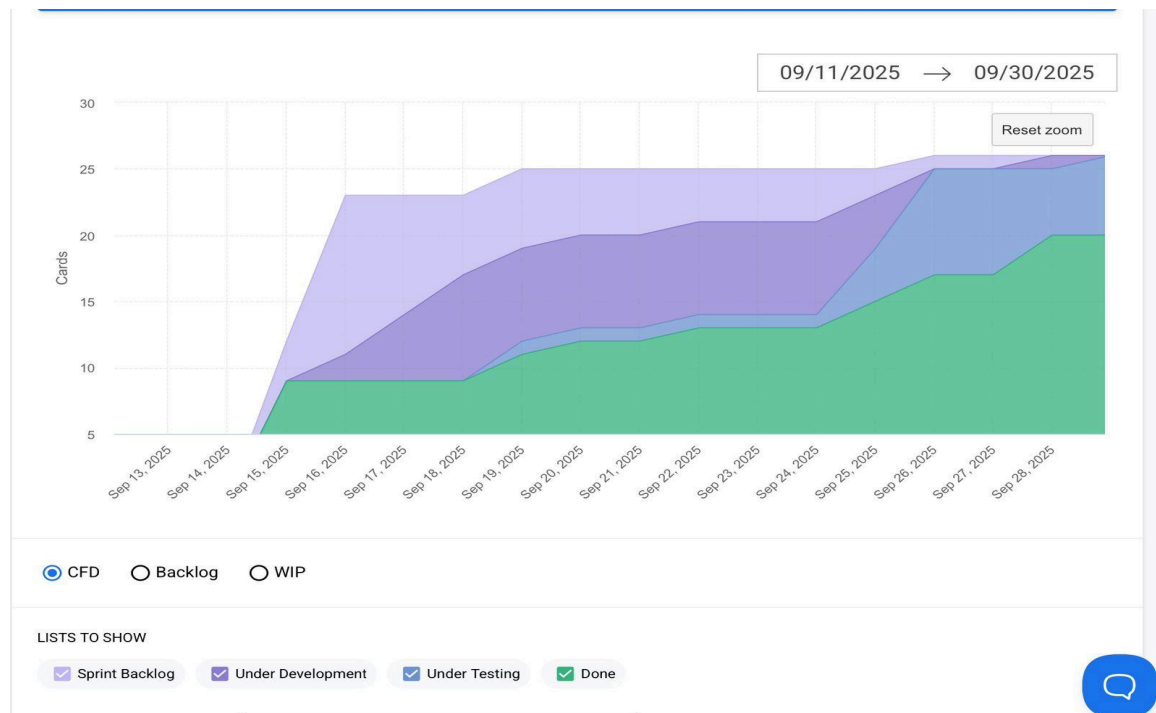


Figure [13]: Sprint 1 Cumulative Chart



Figure [14]: Sprint 2 Cumulative Chart

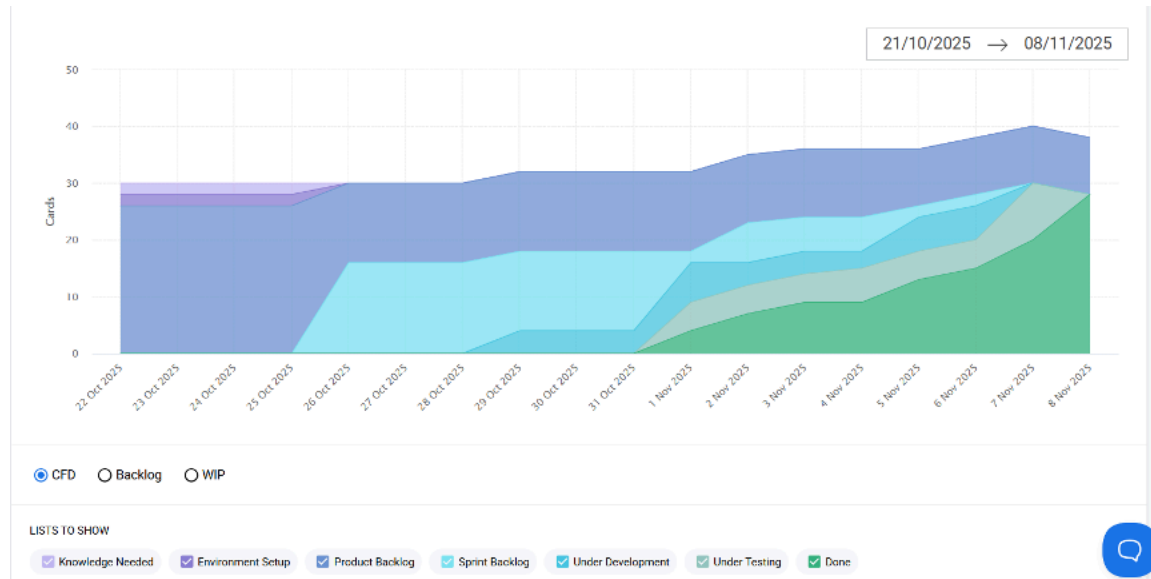


Figure [15]: Sprint 3 Cumulative Chart

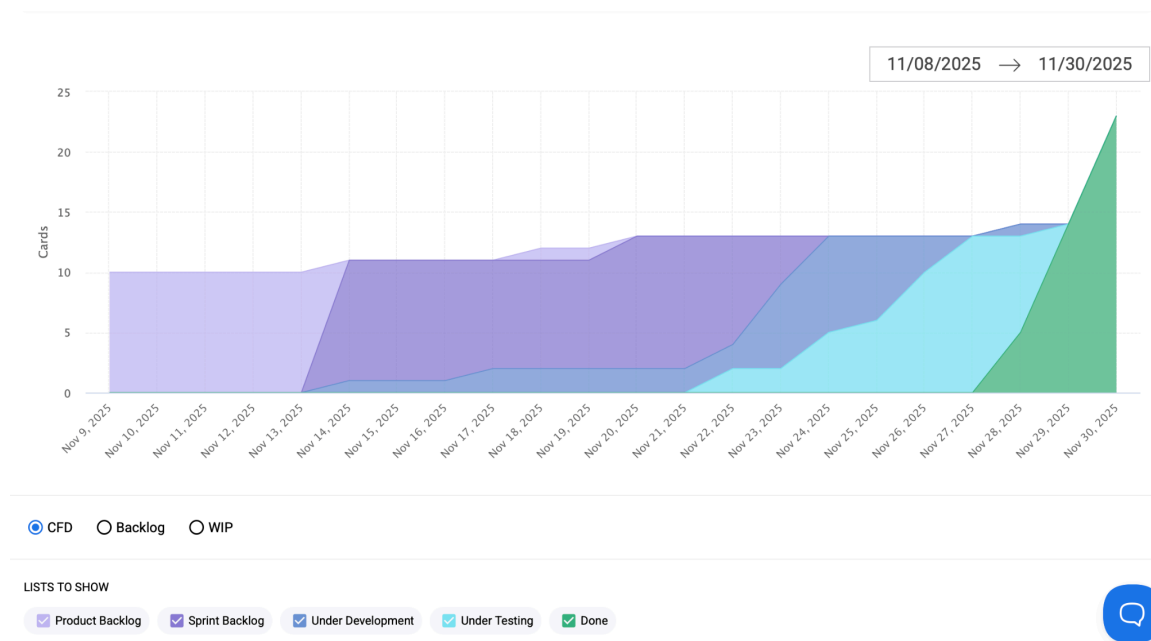


Figure [16]: Sprint 4 Cumulative Chart

The Cumulative Flow Diagram (Nov 9 – Nov 30, 2025) illustrates the flow of work items (Cards) through the five stages: Product Backlog, Sprint Backlog, Under Development, Under Testing, and Done. Initially, from November 9th to November 13th, the total backlog remains stable at 10 cards while work rapidly moves into Under Development, reaching about 10 cards by mid November, with Under Testing and Done remaining at zero. In the mid sprint phase (Nov 15 – Nov 27), the combined Work In Progress (Under Development and Under Testing) stabilizes around 13 cards, indicating a controlled flow, with tasks beginning to transition into Under Testing around November 17th. Done cards start appearing gradually from November 27rd onward. The most significant progress occurs in

the final phase (Nov 28 – Nov 30), where the Done section shows a steep rise, culminating in approximately 23 completed cards by November 30th. This sharp vertical climb confirms a high completion rate and successful delivery of almost all work items, reflecting a strong, healthy flow and a successful close to the tracked period.

Burndown charts

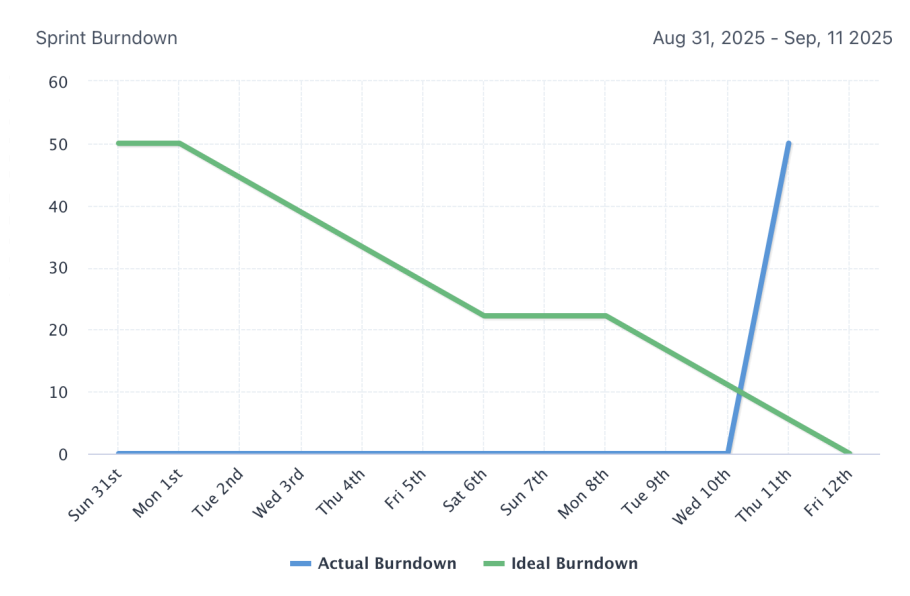


Figure [17]: Sprint 0 Burndown Chart

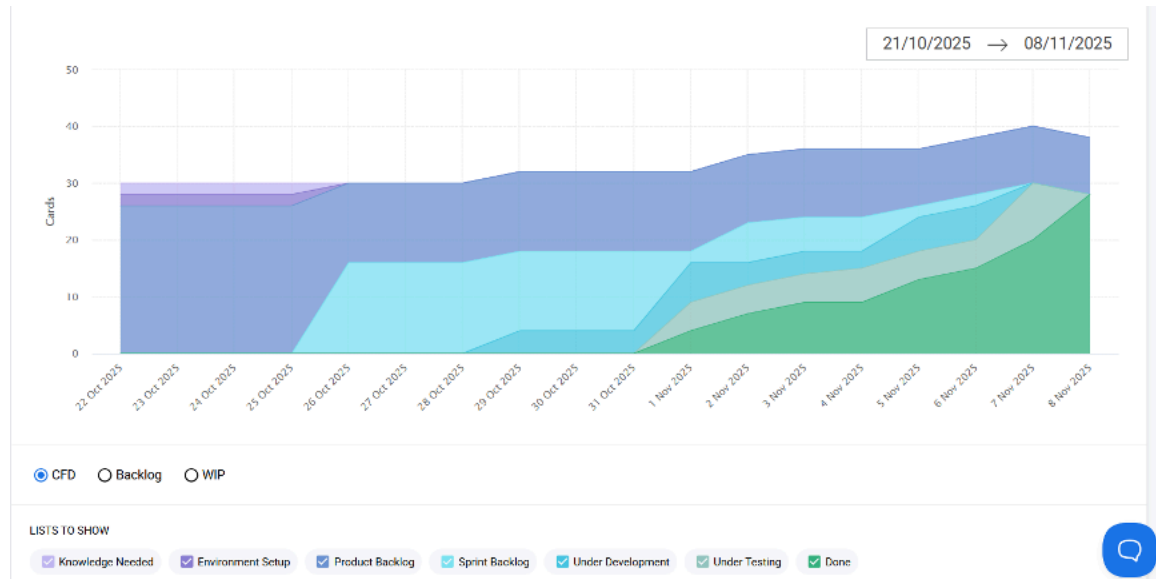


Figure [18]: Sprint 1 Burndown Chart

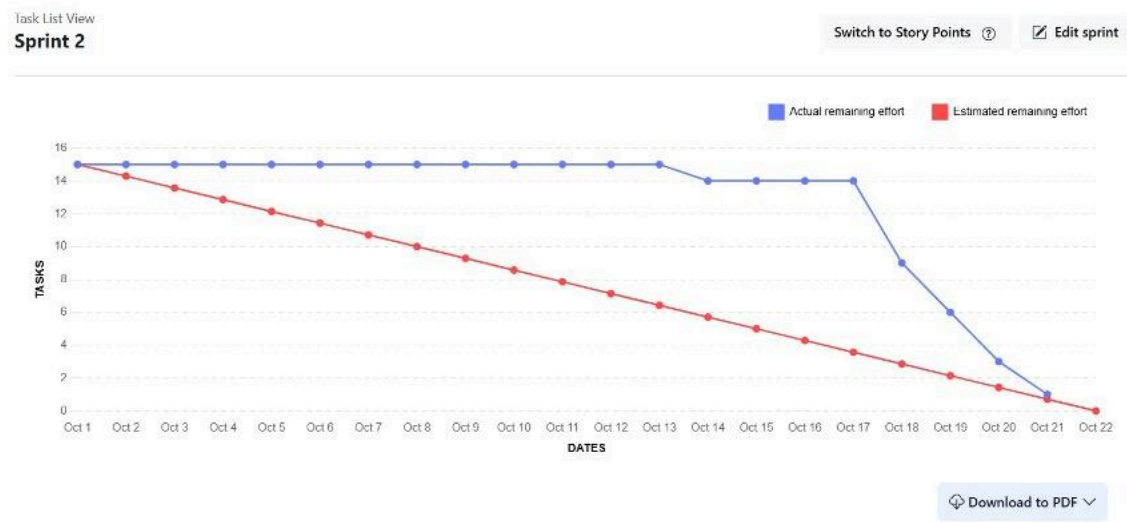


Figure [19]: Sprint 2 Burndown Chart

Burndown chart

Sprint 3

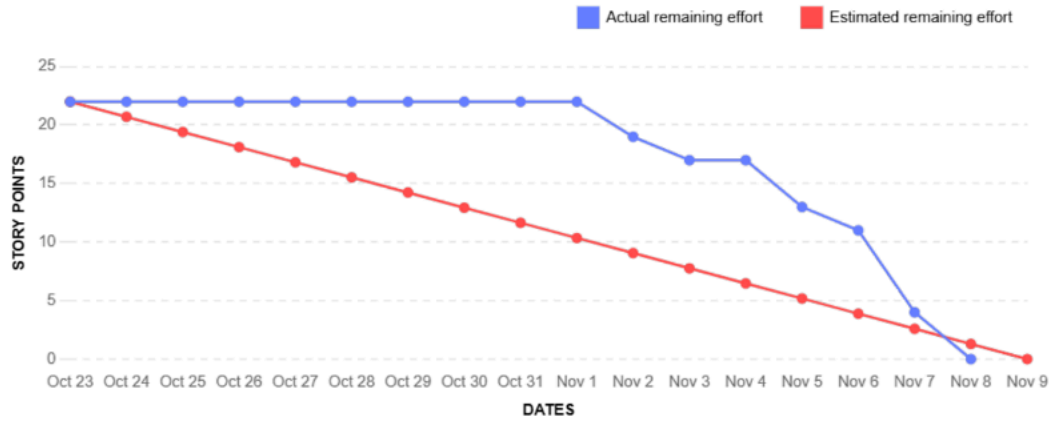


Figure [20]: Sprint 3 Burndown Chart

Sprint Burndown

Nov 09, 2025 - Nov, 30 2025

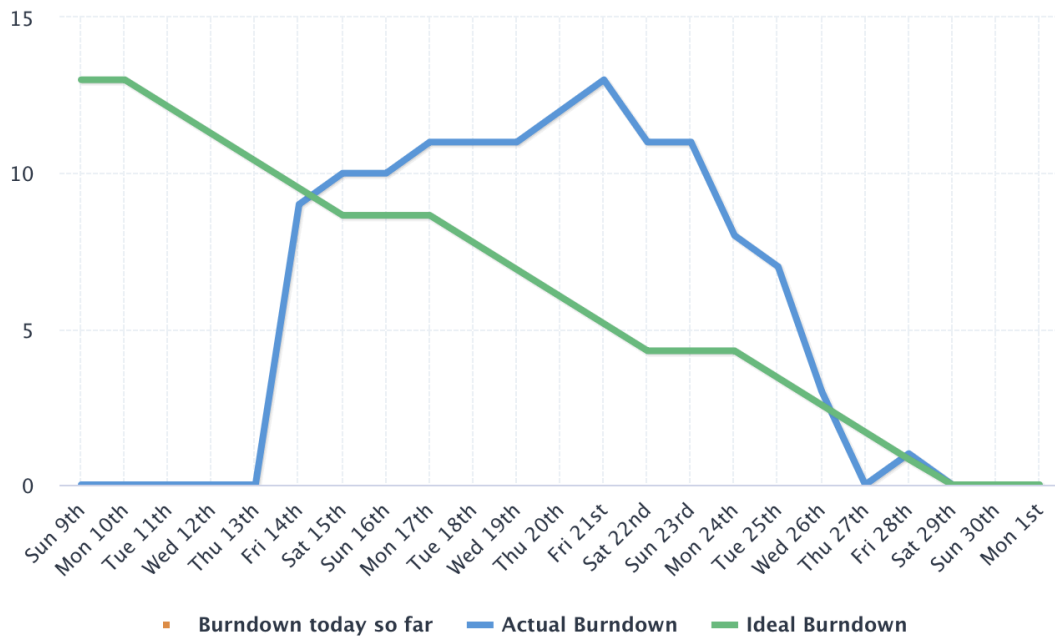


Figure [21]: Sprint 4 Burndown Chart

The Burndown Chart (Nov 09 – Nov 30, 2025) provides a clear view of the team's progress during the sprint. At the start of the sprint, the actual burndown line remained flat, indicating that no

measurable work was completed in the early days. This suggests that tasks were still being set up or were in progress but had not yet reached the “done” stage.

Midway through the sprint, the chart shows a sharp rise in the blue line, reflecting an increase in total story points. This happened because new work or updated requirements were added to the sprint backlog. Immediately after this adjustment, the actual burndown remained steady for a short period, demonstrating that the team was still working through the expanded workload. During the second half of the sprint, the team’s progress accelerated significantly. The blue line begins to drop progressively as more tasks are completed, showing consistent movement toward reducing the remaining effort. A particularly steep decline appears near the final days, indicating that a large portion of the work was completed in a short timeframe.

By the end of the sprint, the actual burndown line reached zero slightly ahead of the ideal line, meaning all committed work including the added items was finished before the sprint ended. This reflects strong team performance, effective task closure, and efficient handling of changes introduced during the sprint.

7. Quality Assurance Activates

7.1 Testing

Our unit testing will cover the Trainee playing mini game story :

“As a Trainee, I would like to play a word-unscramble mini-game where I fix scrambled words by rearranging their letters.”

We will test different cases of answering the game, in this [Testing link](#).

7.2 Version Control- git hub

In version control, we used GitHub to track every change made to the code. This helped us see who changed what and when it happened. Using version control keeps the project organized, makes teamwork easier, and reduces mistakes. It also helps us fix issues faster and review updates more smoothly. Overall, version control helps us deliver high quality work in each sprint because it keeps everything clear, consistent, and easy to manage. [see Appendix 4]

7.3 Checklist

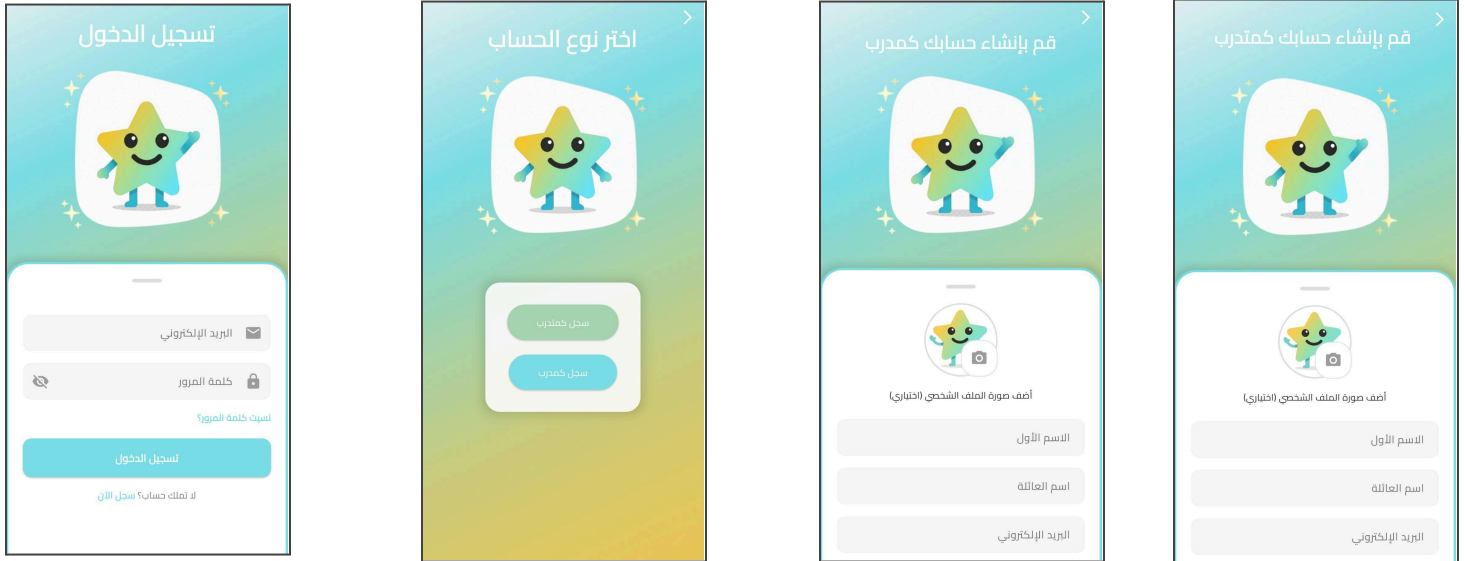
As part of our Quality Assurance practices during this project, we utilized a Checklist to ensure that all functional requirements were properly tested and verified. The checklist included key items such as session management features, rating and reporting functionalities. Each item on the checklist was reviewed and marked as completed once verified, providing a clear and systematic way to track testing coverage. Using a checklist helped our team identify any gaps or issues early, ensured consistency in testing procedures, and served as documented evidence of quality control. [see Appendix 7]

7.4 Pair Programming

As part of our Quality Assurance practices, we applied Pair Programming throughout this project. Usually, two of us would meet on Microsoft Teams to code together, fix errors, and test the runs in real time. If one of us faced an issue or the code didn't run properly, we worked through it side by side until it was resolved. This collaborative approach helped us detect mistakes early, share knowledge efficiently, and keep our code consistent across the project. Working in pairs also strengthens communication and made development smoother, ensuring higher quality results for each feature we delivered. [see Appendix 6]

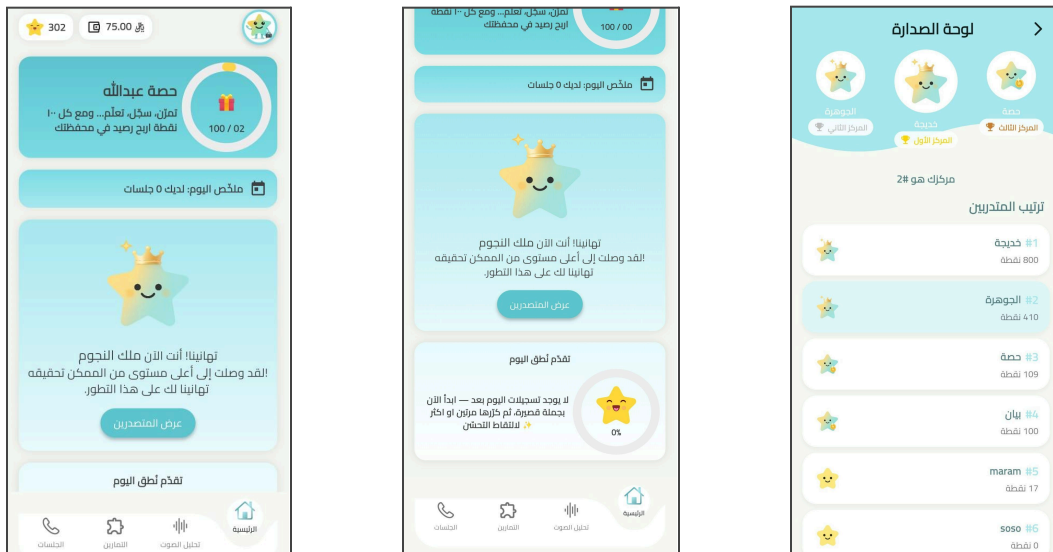
8. System Screenshots

- Sign In, Sign up Specialist, Sign up Trainee.



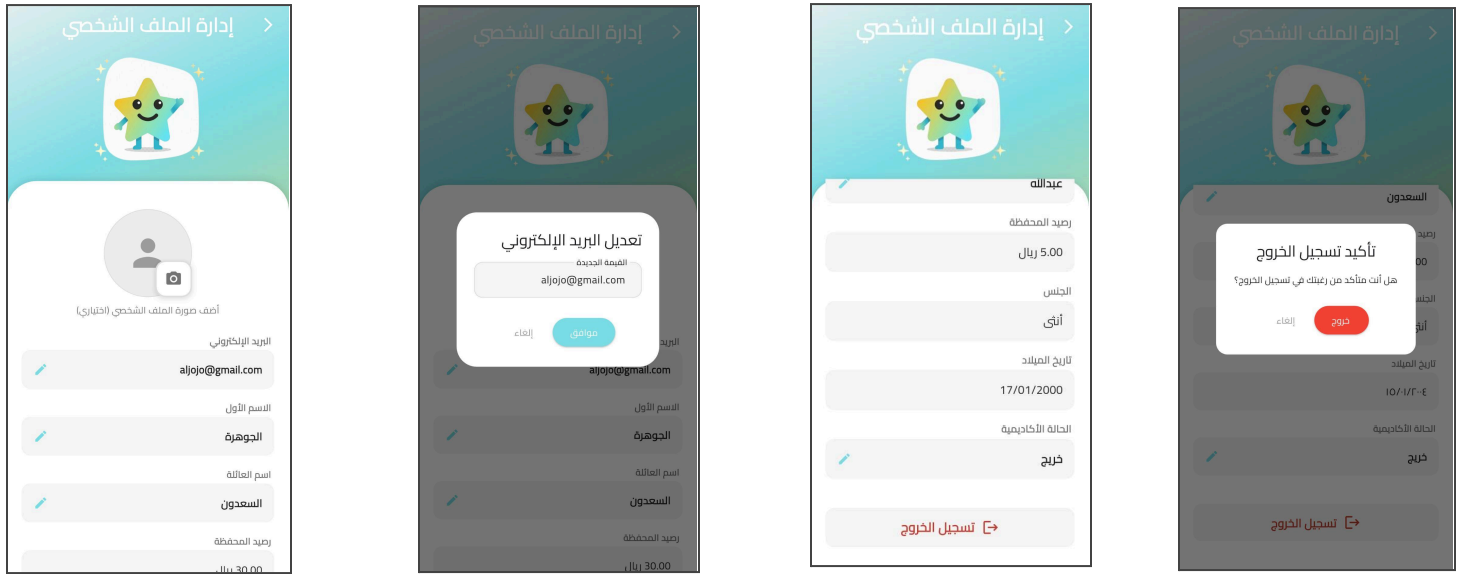
Figure[22]: Sign in, Sign up

- **Trainee Features**
- Trainee's Dashboard and score board



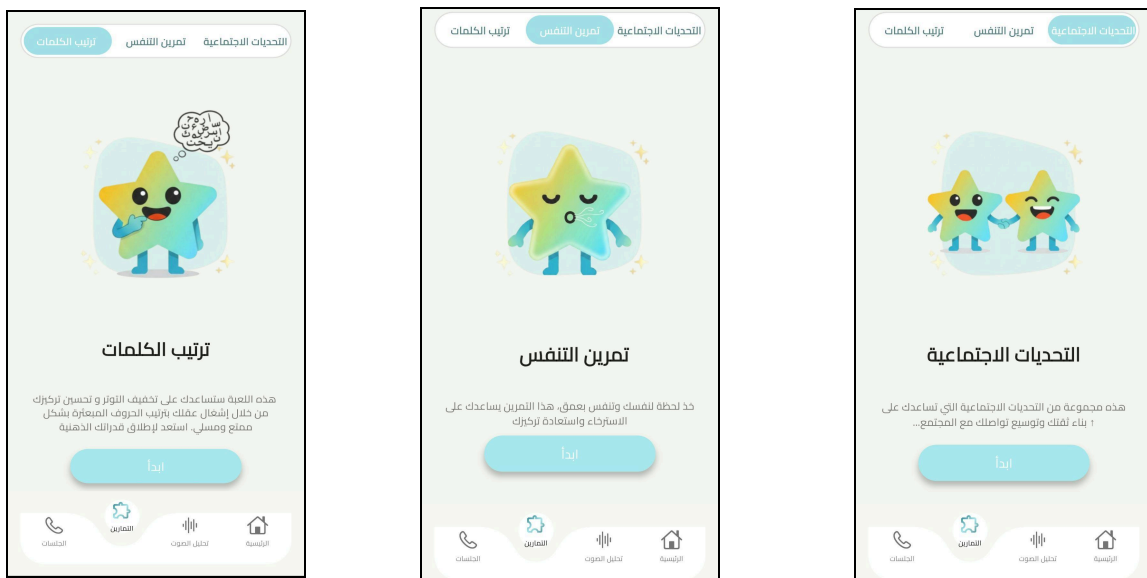
Figure[23]: Trainee Dashboard and score board

- Trainee's Profile



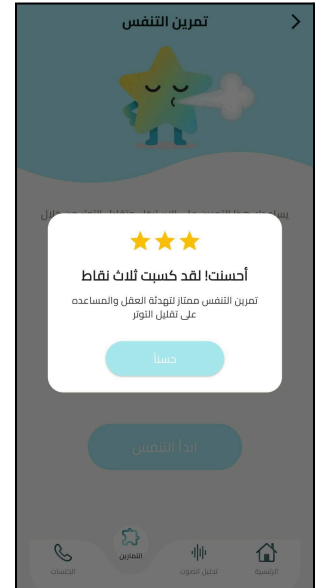
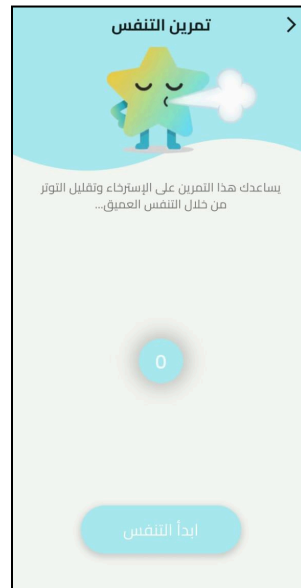
Figure[24]: Trainee's Profile

- Trainee's Activities



Figure[25]: Activities

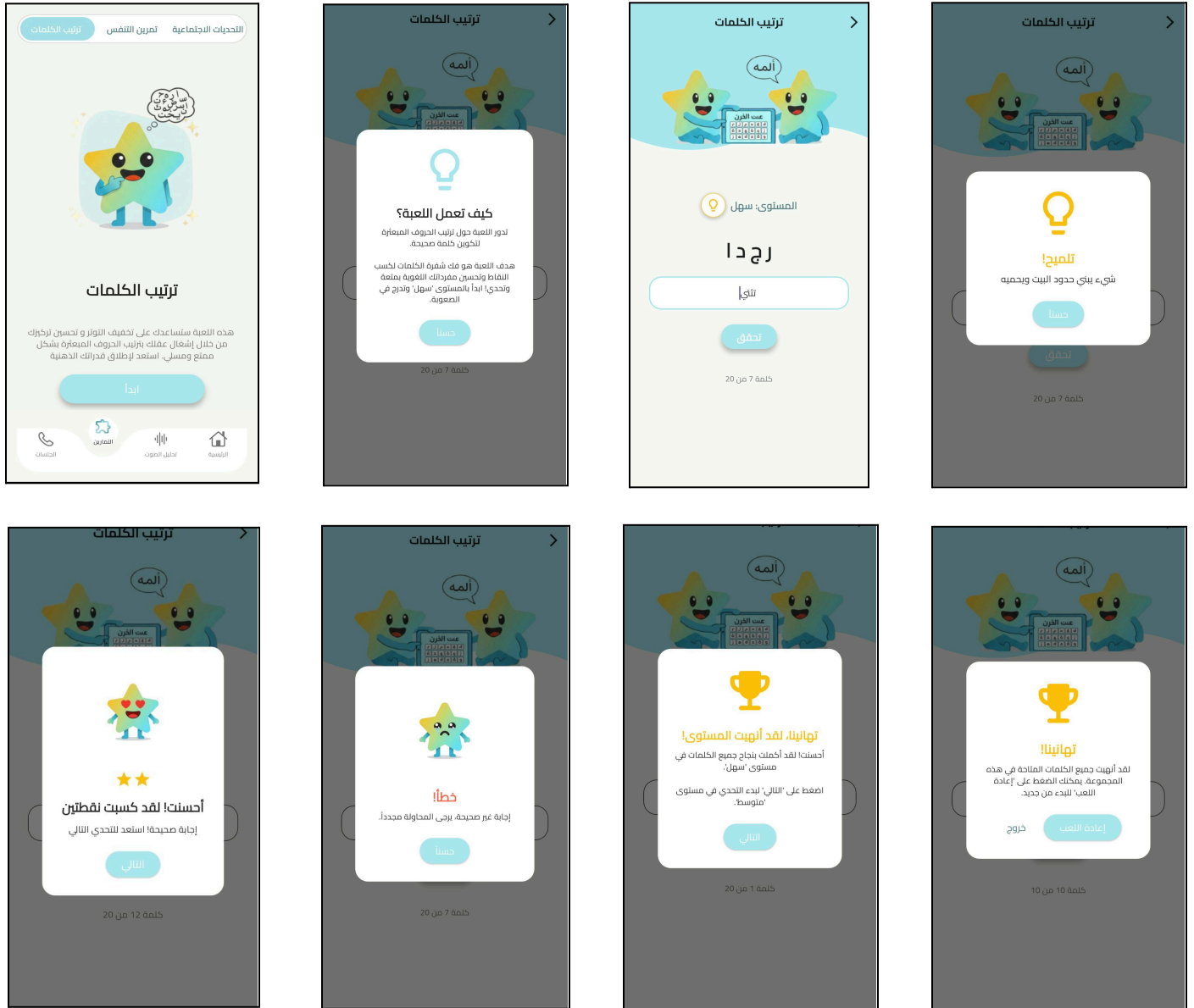
- Trainee Breathing Exercise



:

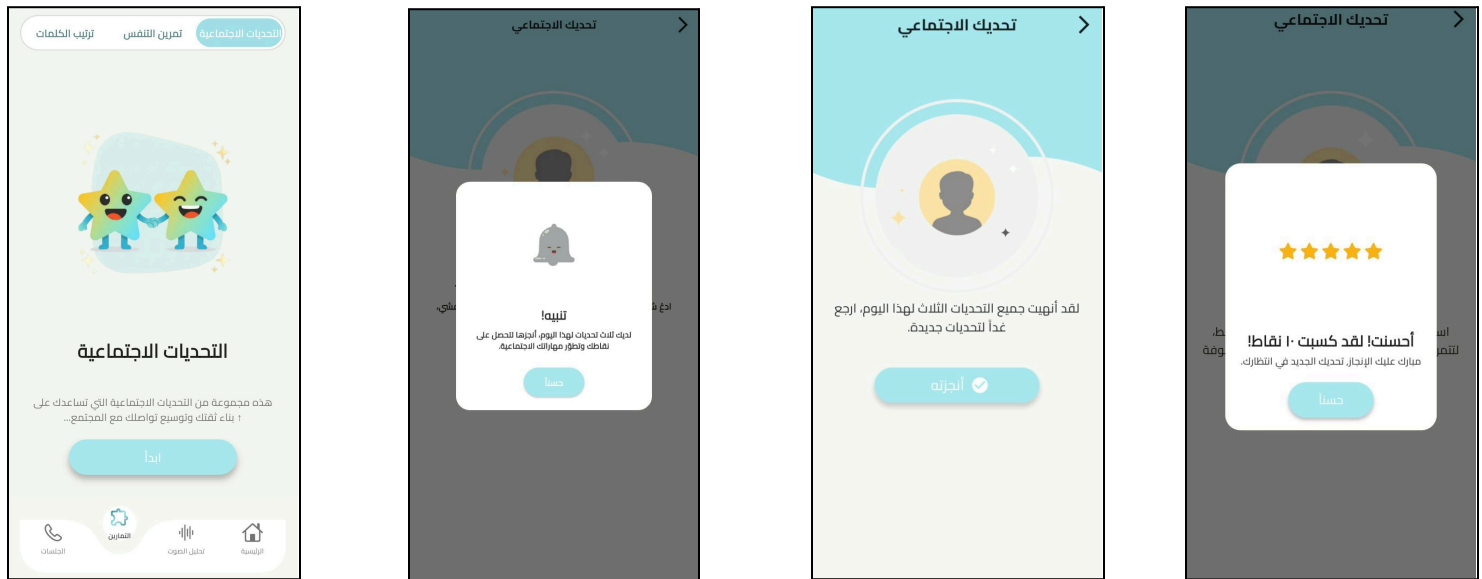
Figure[26] :Breathing exercise

- Trainee word - unscramble game



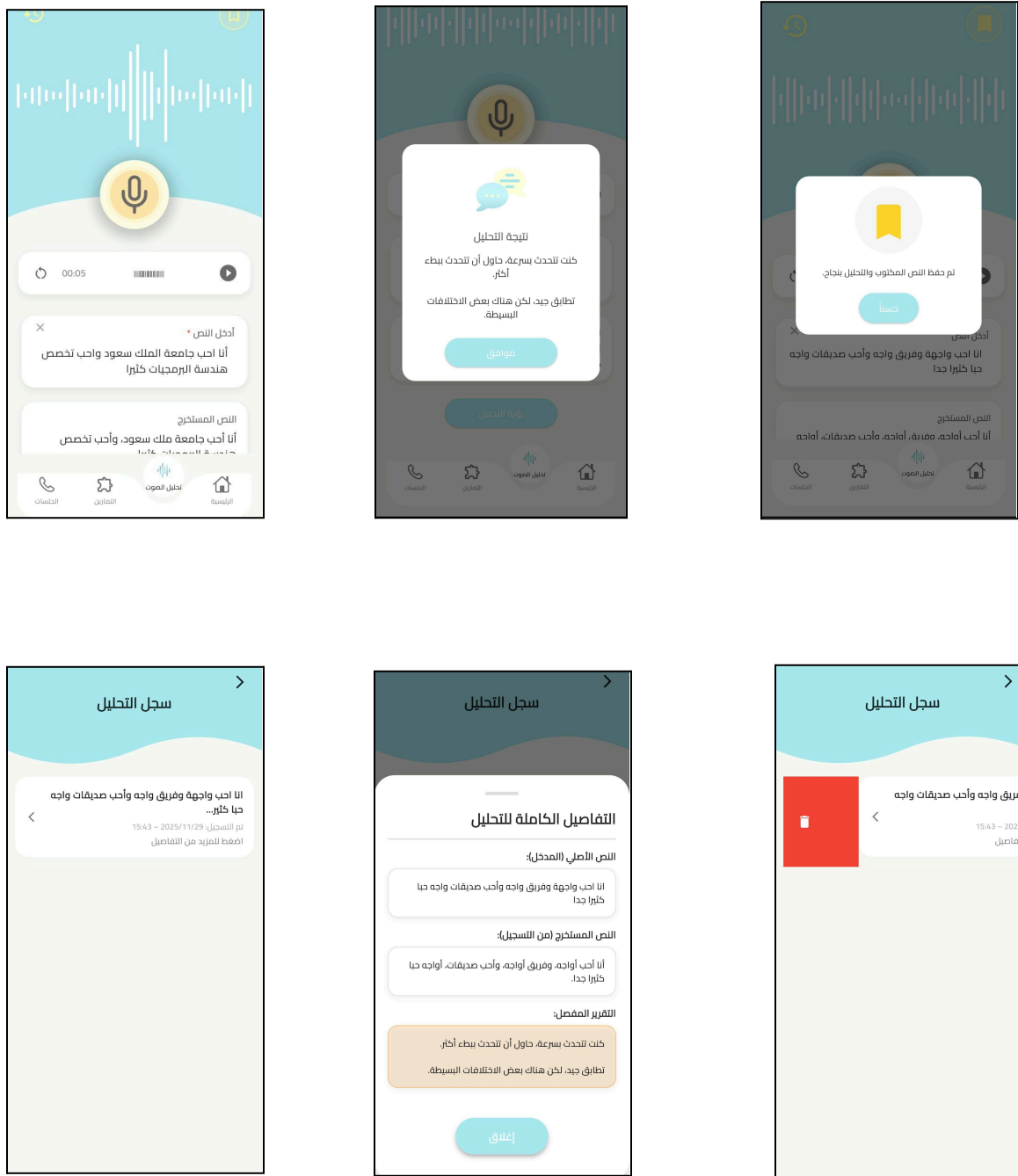
Figure[27]: word - unscramble game

- Trainee Social Challenges



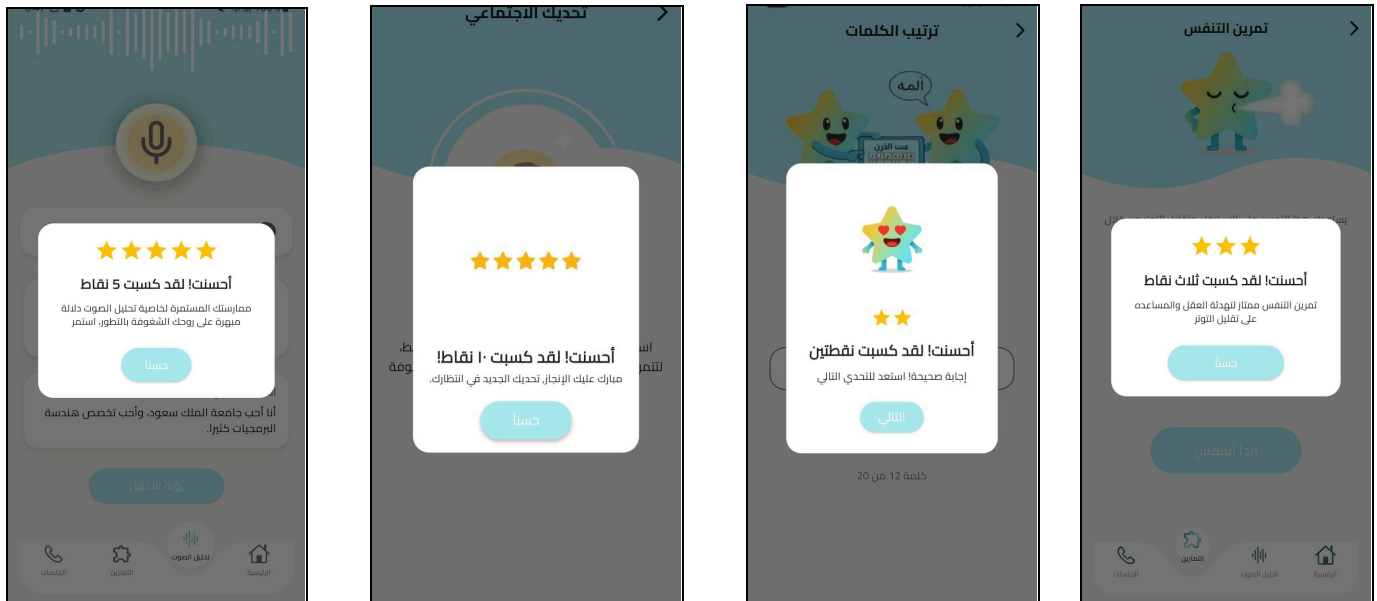
Figure[28]: Social challenges

- Trainee Speech Analysis and save and history



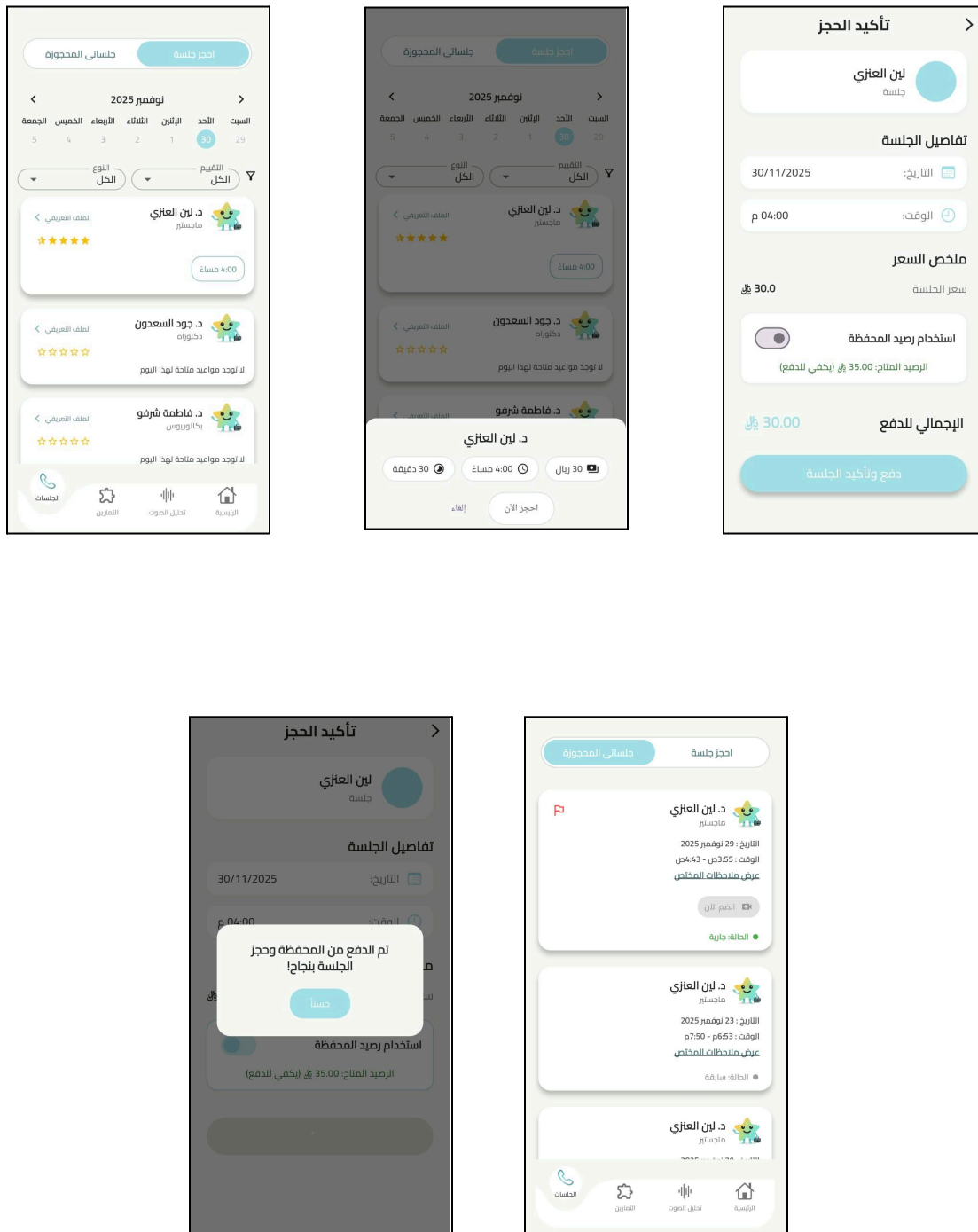
Figure[29]: speech analysis history

- Trainee scores



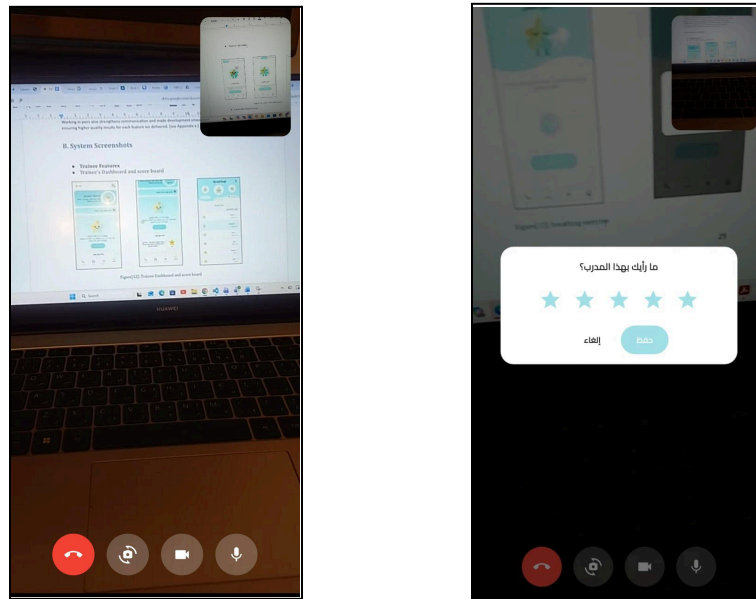
Figure[30]: Scores

- Trainee Book a session with Specialist



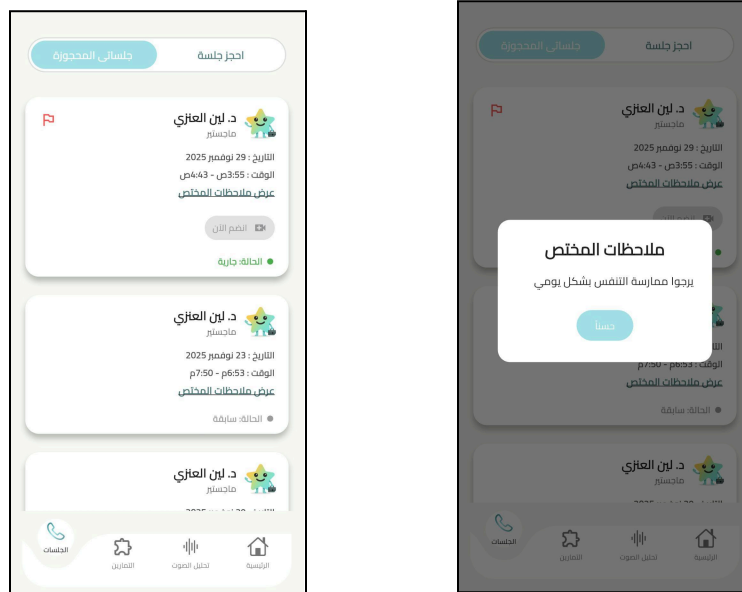
Figure[31]: Book a session

- Trainee join a session, and rate a specialist.



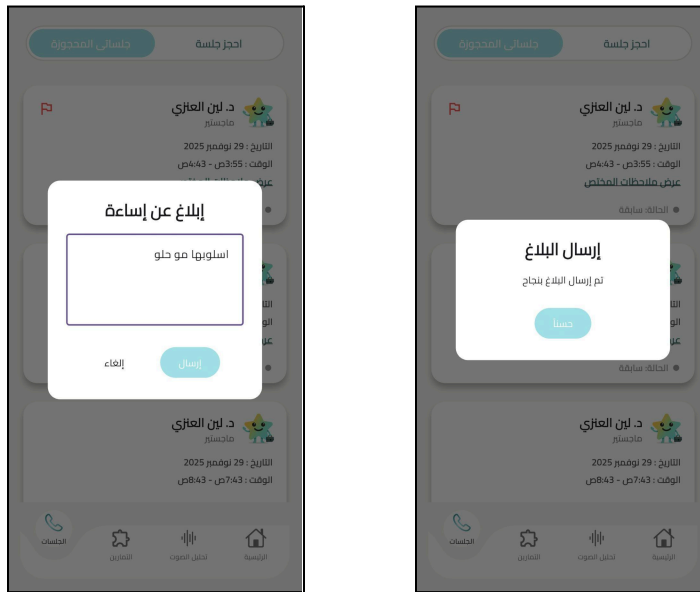
Figure[32]: Join session and rating

Trainee view specialist feedback



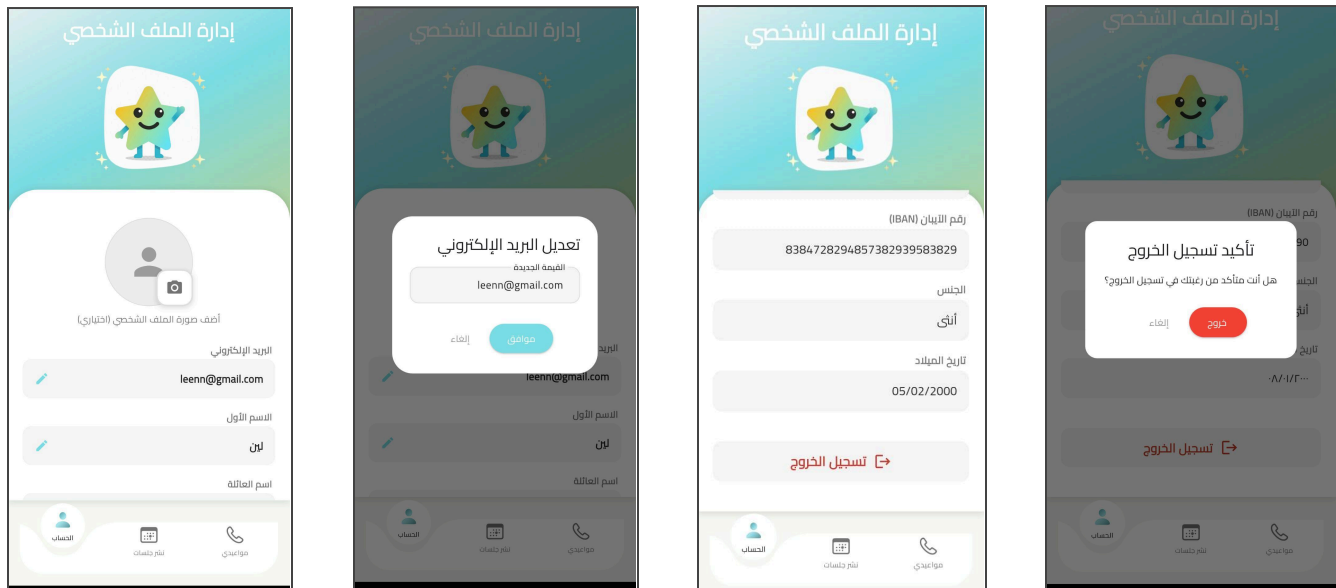
Figure[33]: viewing Specialist feedback

- Trainee Report specialist



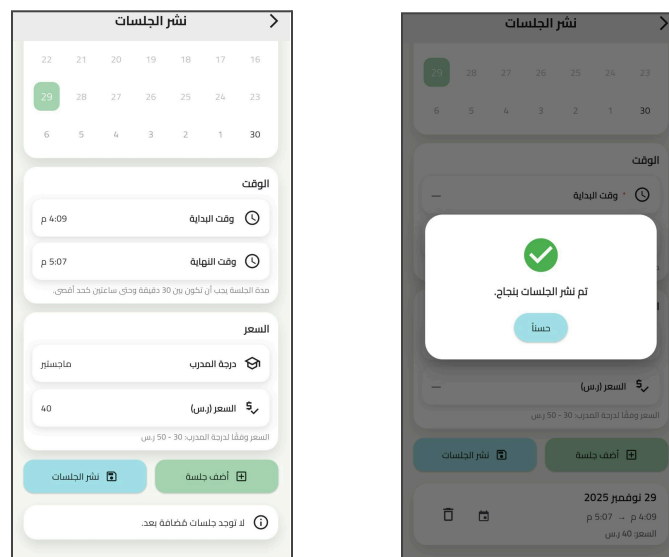
Figure[34]: Report Specialist

- **Specialist Features**
- **Specialist Profile**



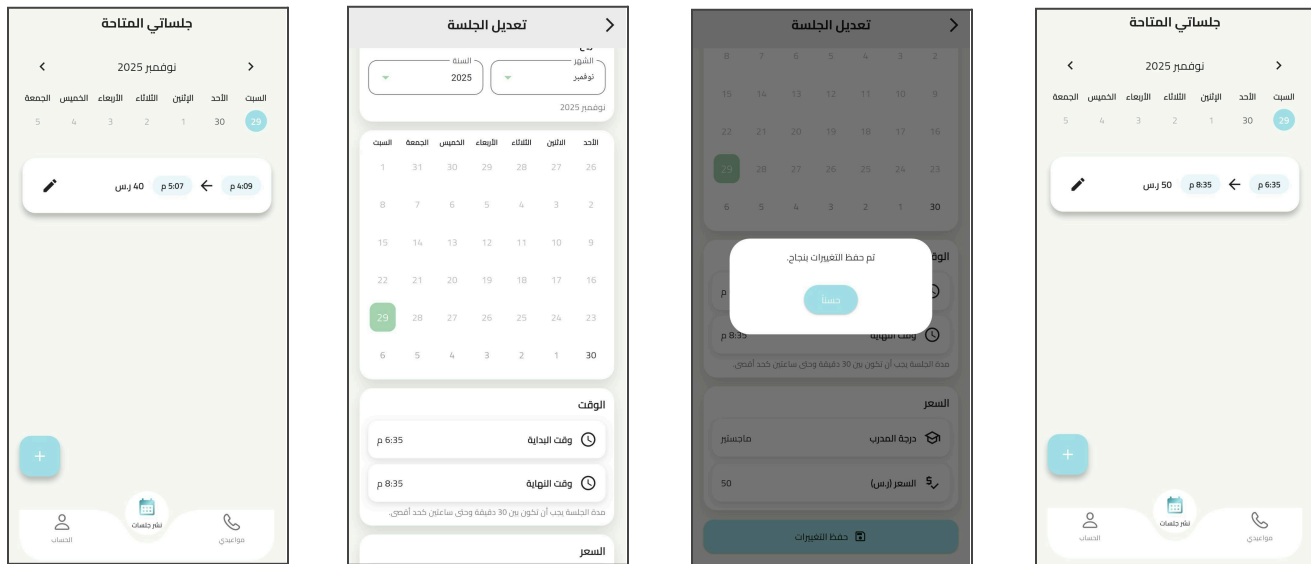
Figure[35]: Specialist Profile

- **Specialist Publish a Session**



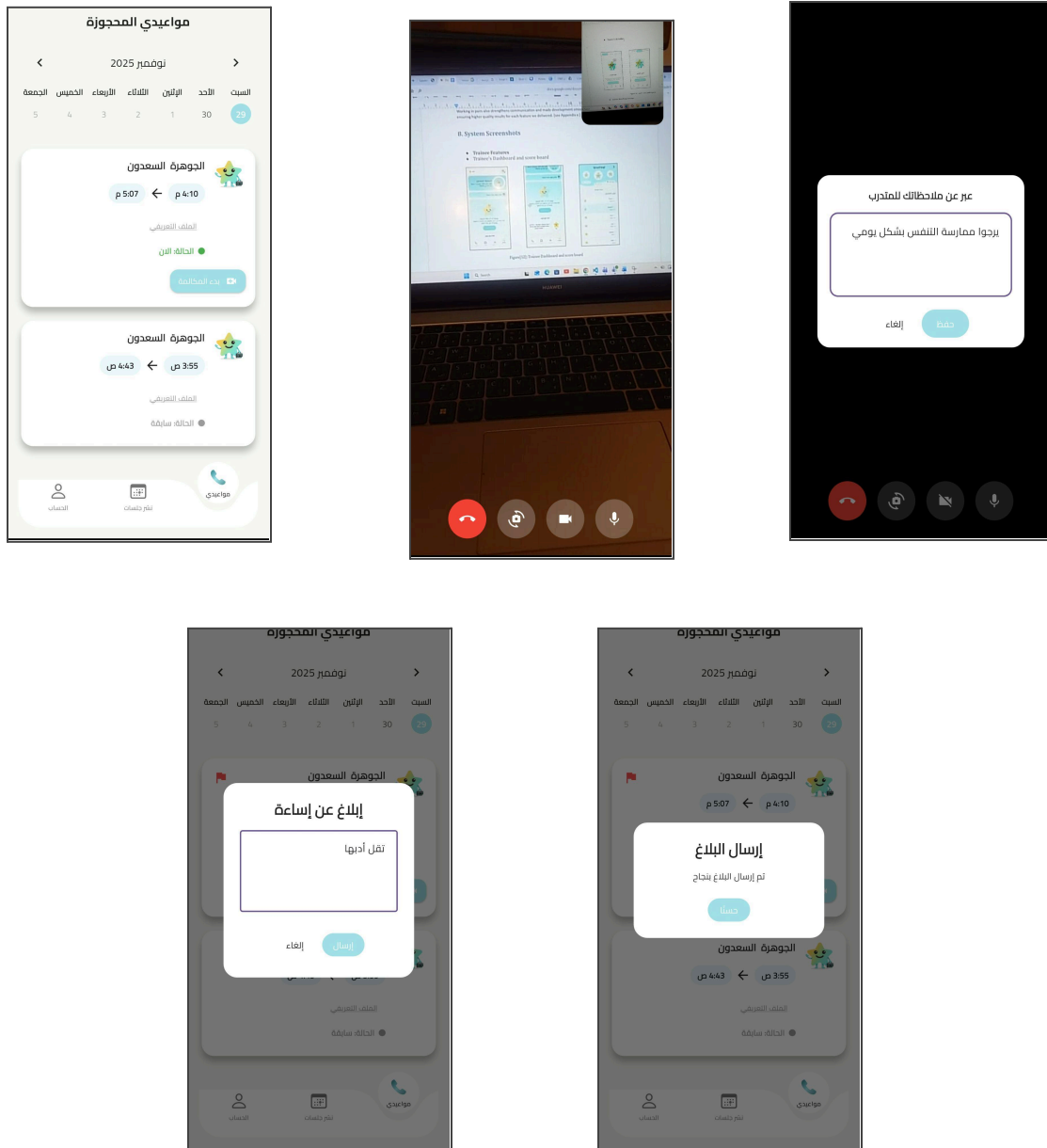
Figure[36]: Publish session

- Specialist Reschedule Session
-



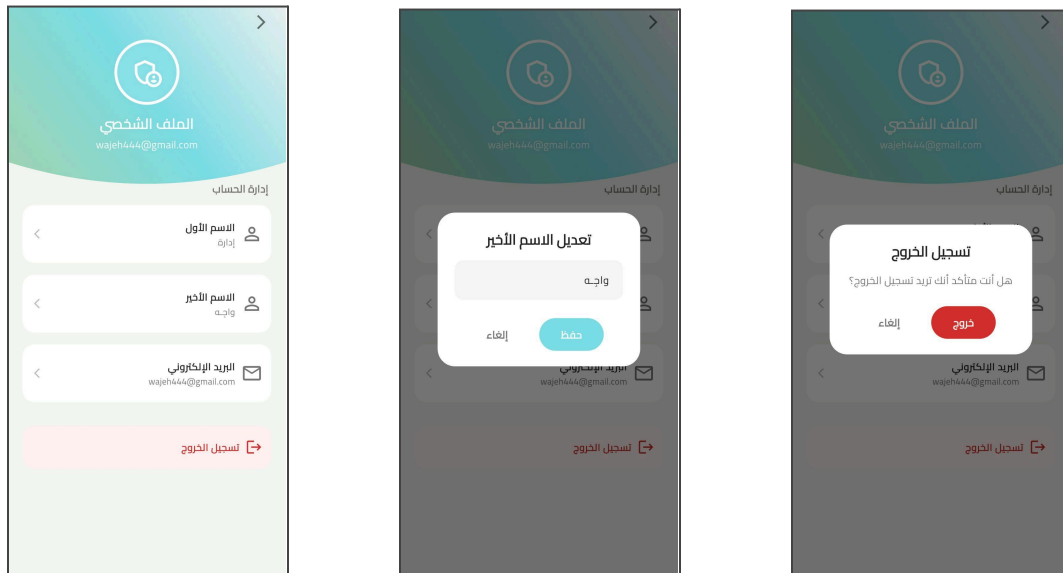
Figure[37]: Reschedule session

- Specialist View scheduled Session,Start Session, add feedback,and report trainee.



Figure[38]: View session,start session, add feedback ,and report

- Admin Features
- Admin Profile



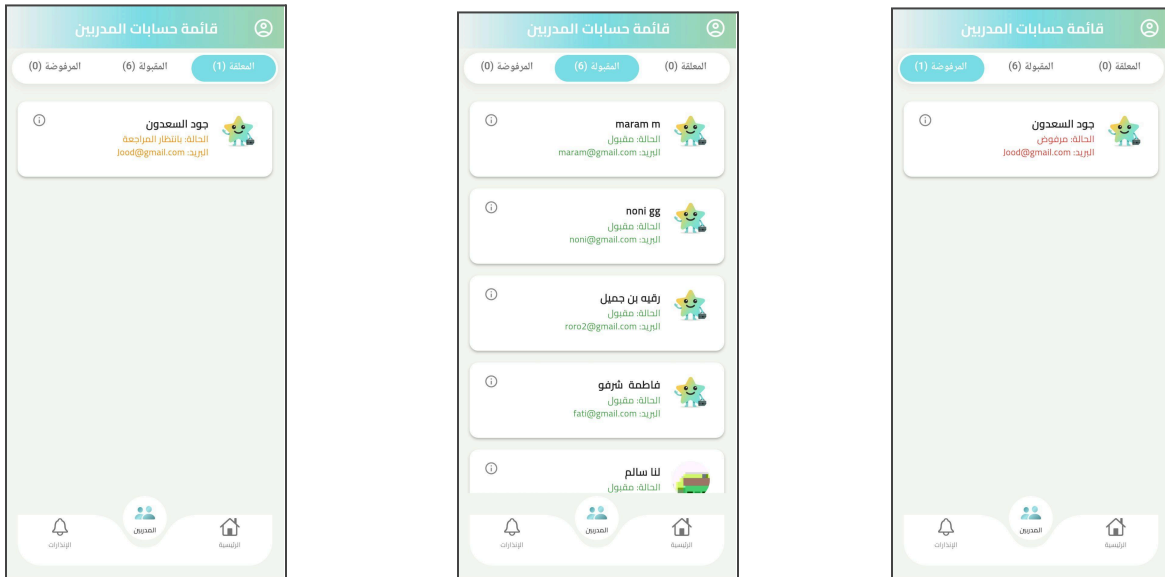
Figure[39]: Admin Profile

- Admin Dashboard



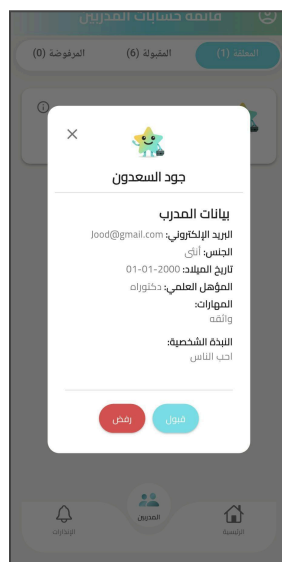
Figure[40]: Admin Dashboard

- Admin Specialist List (Pending,Accepted,Rejected)



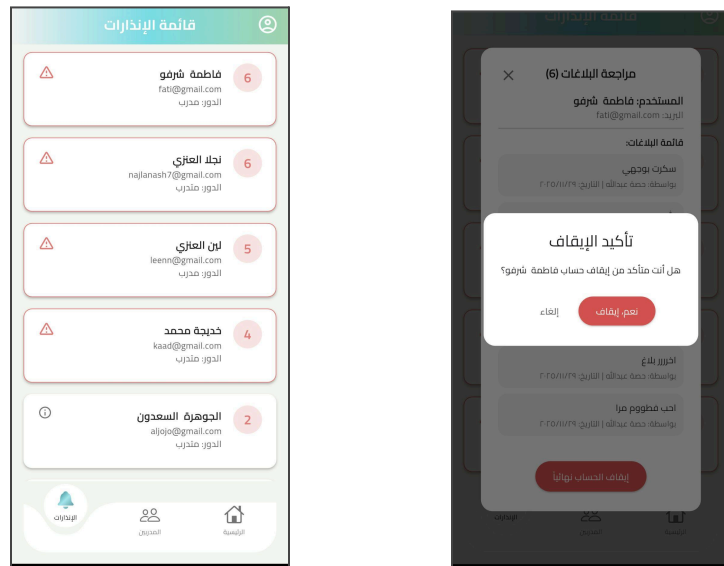
Figure[41]: Specialist List

- Specialist Request Information



Figure[42]: Specialist request information

- Report List and block specialist



Figure[43]: Report List, block specialist

9. Acquired Knowledge

During the lifecycle of the Wajeh project, our team gained significant insights encompassing project management, team collaboration and technical proficiency:

Project management skills:

- **Team collaboration:** Learned how to work effectively as a team holding regular meetings and defining everyone's role to make sure we are all on the same page.
- **Problem Solving:** Improved our ability to handle unexpected technical issues and find solutions without delaying the project progress.
- **Requirements Prioritization:** we applied the MoSCoW framework to filter project requirements. This approach allowed us to distinguish between cortical deliverables and optional enhancements, ensuring to dedicate our effort to high impact features.

Technical skills:

- **App development stack:** We gained practical experience with Flutter and Android Studio, learning how to structure a full mobile app workflow and integrate tools effectively to build user-centric features.
- **Data & Auth Management:** We developed a stronger understanding of managing real-time data through Firebase Realtime Database and learned how secure authentication systems are designed and implemented.
- **API connectivity:** By integrating Azure (Speech Analysis), Agora (Video Calls), and Stripe (Payments), we enhanced our ability to work with external services and understood how APIs expand an app's functionality.
- **Programming Skill Growth:** Our skills in Dart improved significantly as we learned to write more optimized, maintainable, and scalable code aligned with real application requirements.
- **Source Code Control:** Using GitHub taught us how to collaborate efficiently, resolve merge conflicts, track changes properly, and maintain an organized codebase in a team environment.

UI/UX Design Implementation:

- **User centric interface:** We learned how to think from the user's perspective by designing intuitive UI flows, selecting suitable Flutter widgets, and improving the overall user journey.

Testing & Quality Assurance:

- **Performance Optimization:** We understood how to analyze and enhance app performance, reduce loading times, and ensure the app responds smoothly under different conditions.
- **Testing & Debugging:** Through extensive manual testing, we strengthened our debugging skills and learned systematic methods to identify, document, and fix issues across various Android devices.

10. Future Work

At Wajeh , our goal is to help users improve their communication skills and build confidence. To support that mission even further we plan to introduce additional features that will expand the app's capabilities, make the experience more engaging ,and impactful , and reach a wider community.

1. IOS Support:

Ensure full compatibility across multiple operating systems, on IOS. This allows users to access Wajeh seamlessly on any device, making it more flexible and convenient for different environments.

2. Community chats:

Introducing interactive community chats where users can connect, share experiences, and practice communication in a supportive environment. These will foster collaboration and motivation, helping users feel part of a larger community.

3. Smart , personalized AI Assistant :

A smart AI assistant that guides users through their journey by analyzing their progress and providing personalized exercises, tips and feedback. It will adapt to each user's strengths and weaknesses, creating a tailored experience that helps improvement and keep users engaged .

4. More Exercises :

expand the app's library with diverse exercises targeting different aspects of communication and social skills. Which will allow users to practice in varied contexts to develop a wider range of skills.

5. Multi-language Support :

Support multiple languages to make Wajeh accessible to boarder audience. This benefits not only foreign communities but also users who hesitate to speak in non native languages, providing guidance and practice in a comfortable and supportive way.

6. Enhanced Voice Analysis :

Upgrade the voice analysis feature using advanced AI algorithms for real time audio and video conversations and feedback on speech clarity, tone, pace, and confidence. Which will be even more powerful than just a voice record.

11. Conclusion

Wajeh is an android mobile application designed to help individuals to overcome communication barriers, manage anxiety, and enhance their speaking abilities in various social, academic and professional situations. By combining guided and gamified exercises, live training sessions with certified specialists, social challenges, and intelligent speech analysis, the app provides a structured path for personal and social improvement. through features like progress tracking, and points based reward system, the platform offers a Supportive and motivating environment for consistent practice. Wajeh is a space where every one can communicate as if they carry absolute confidence in their speech.

12. References

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13. Appendices

1.Stripe API

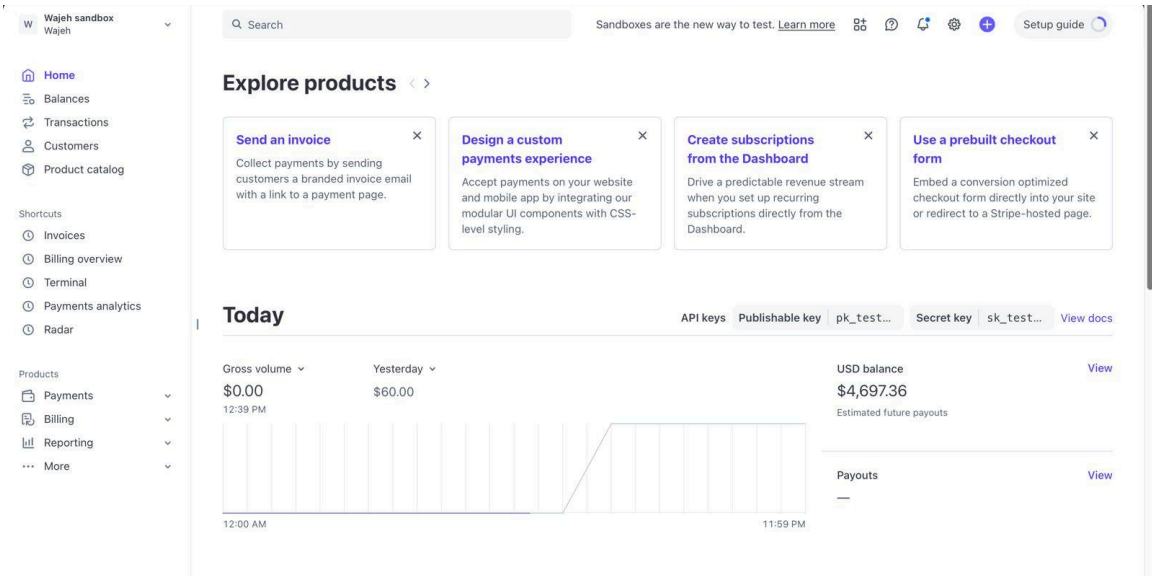
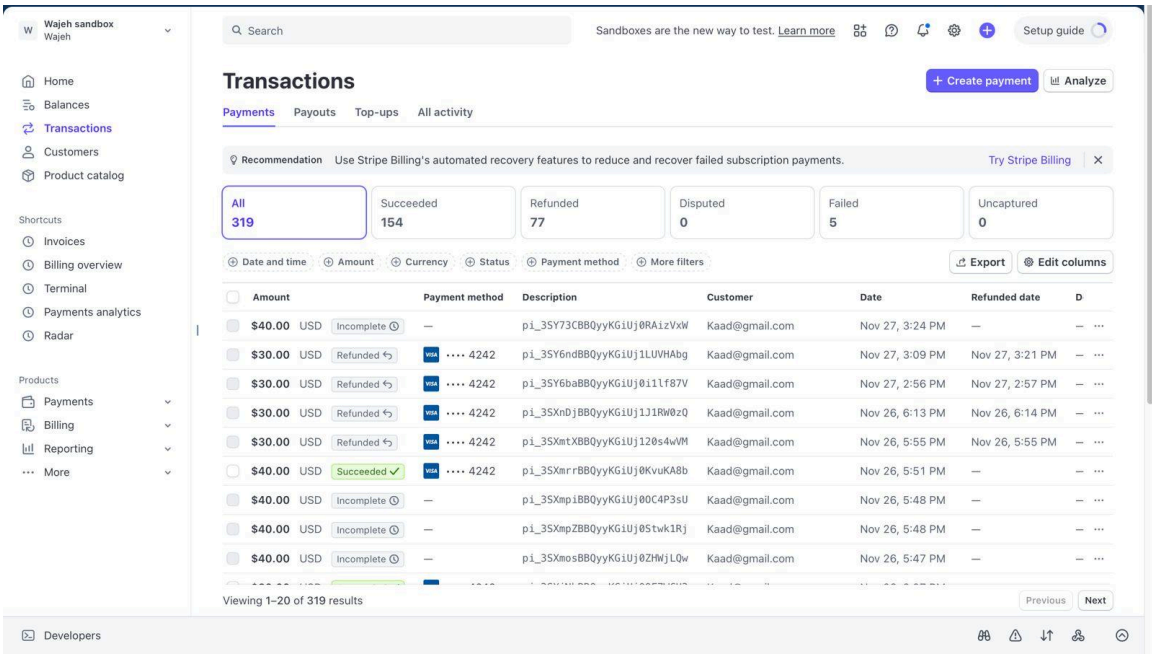
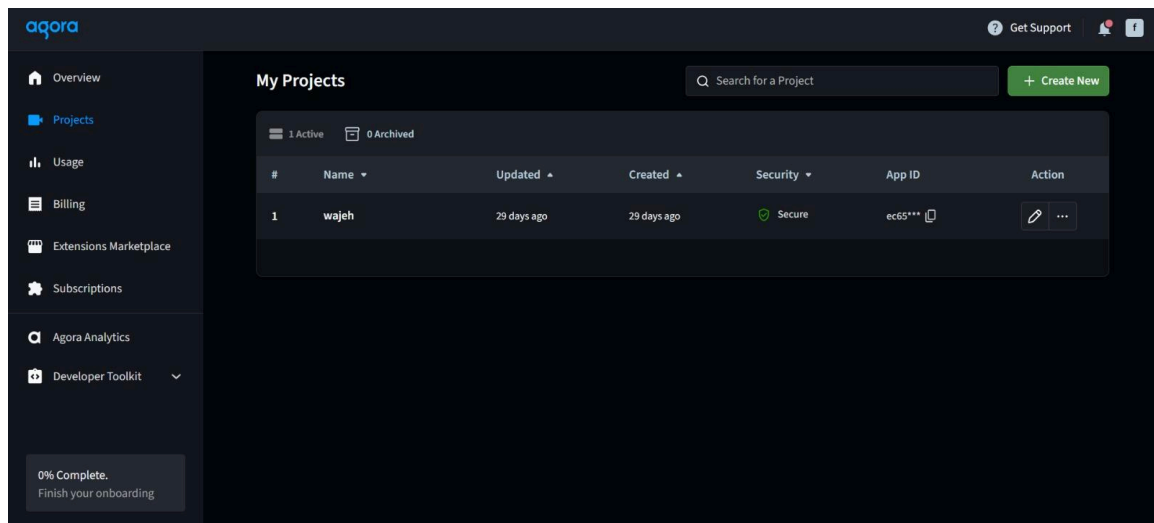


Figure [44]: Payment processing flow integrated via Stripe API

2.Agora API



Date/Time	Audio	Video HD	Video FHD	Video 2K	Video 2K+
2025-10-30	5.47	0	0	0	0
2025-10-31	10.77	3.15	0	0	0
2025-11-01	0.48	0.33	0	0	0
2025-11-02	3.82	1.33	0	0	0
2025-11-03	25.03	0.47	0	0	0
2025-11-04	0	0	0	0	0
2025-11-05	0	0	0	0	0
2025-11-06	4.77	1.6	0	0	0
2025-11-07	5.68	0	0	0	0
2025-11-08	1.73	0.22	0	0	0
2025-11-09	1.38	0.38	0	0	0
2025-11-10	0	0	0	0	0
2025-11-11	0	0	0	0	0

0% Complete.
Finish your onboarding

Figure [45]: Real-time communication interface powered by Agora API

3. Azure speech-to-text API

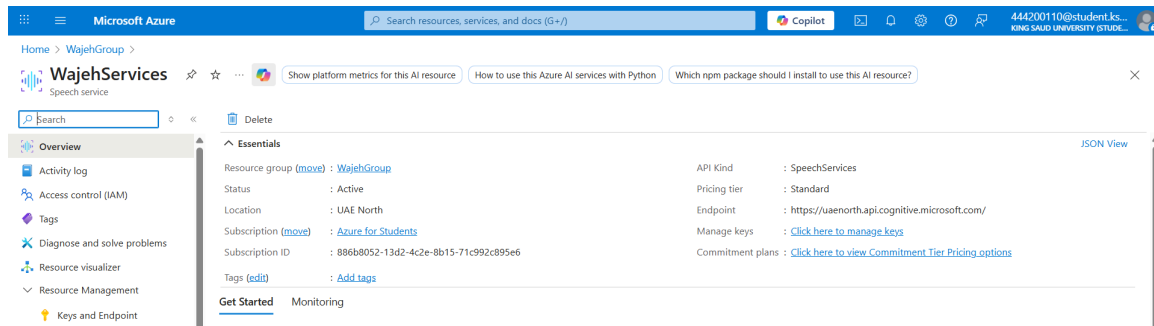


Figure [46]: Azure Speech Services resource configuration used in **WAJEH**

4. GitHub

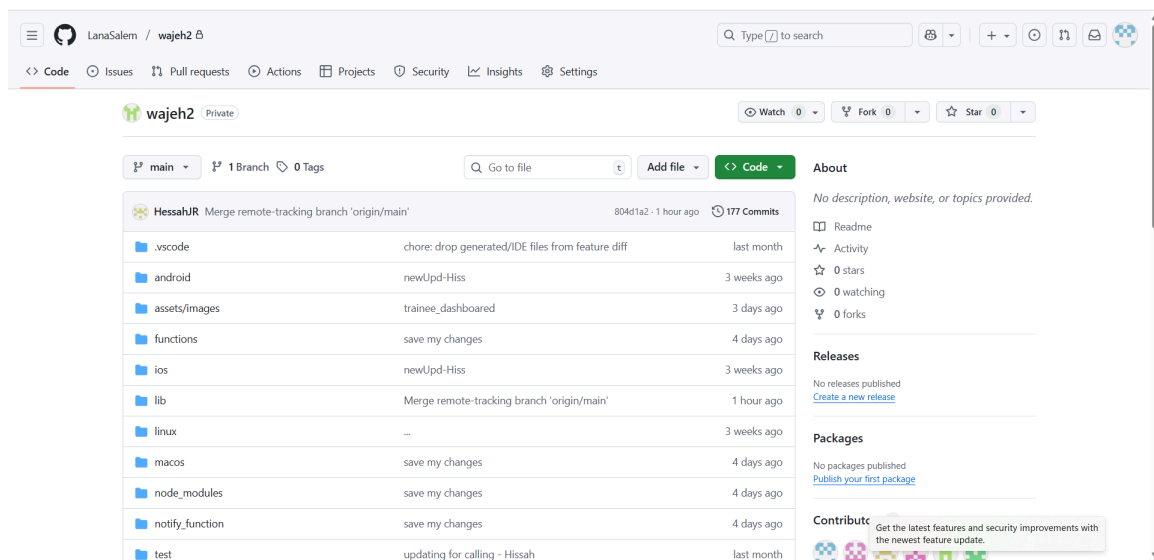


Figure [46]: GitHub repository of **WAJEH** app

5.Trello board

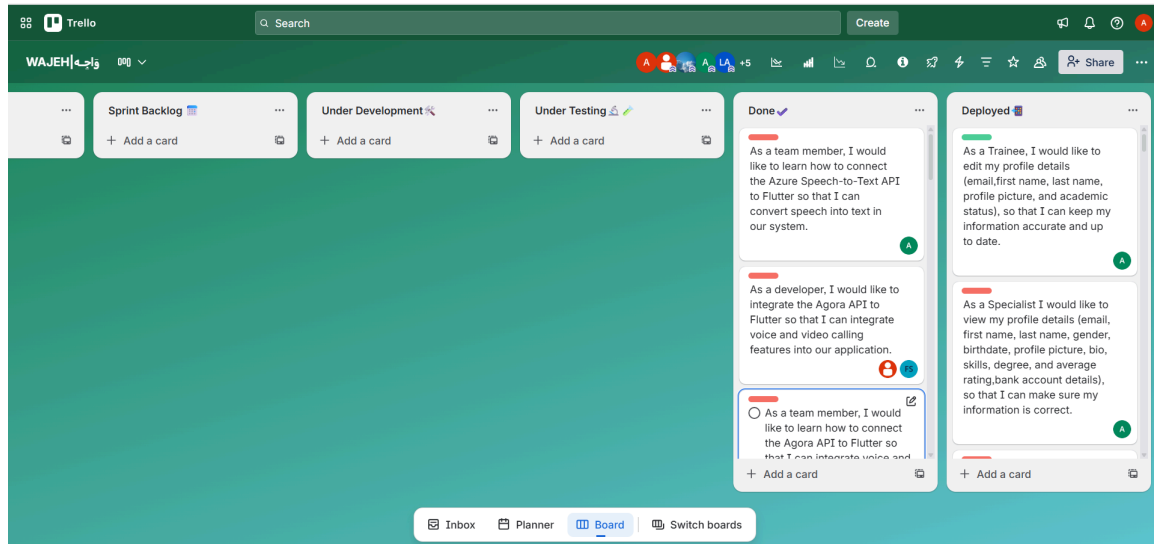


Figure [48]: WAJEH Trello board

6. Pair programming

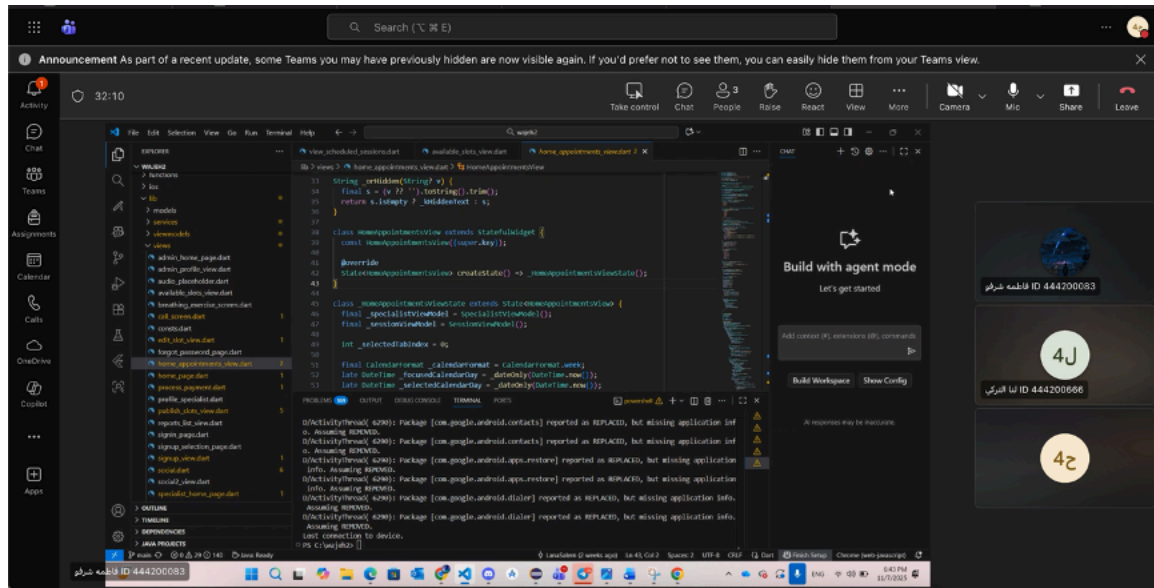


Figure [49]: pair programming

7. Checklists

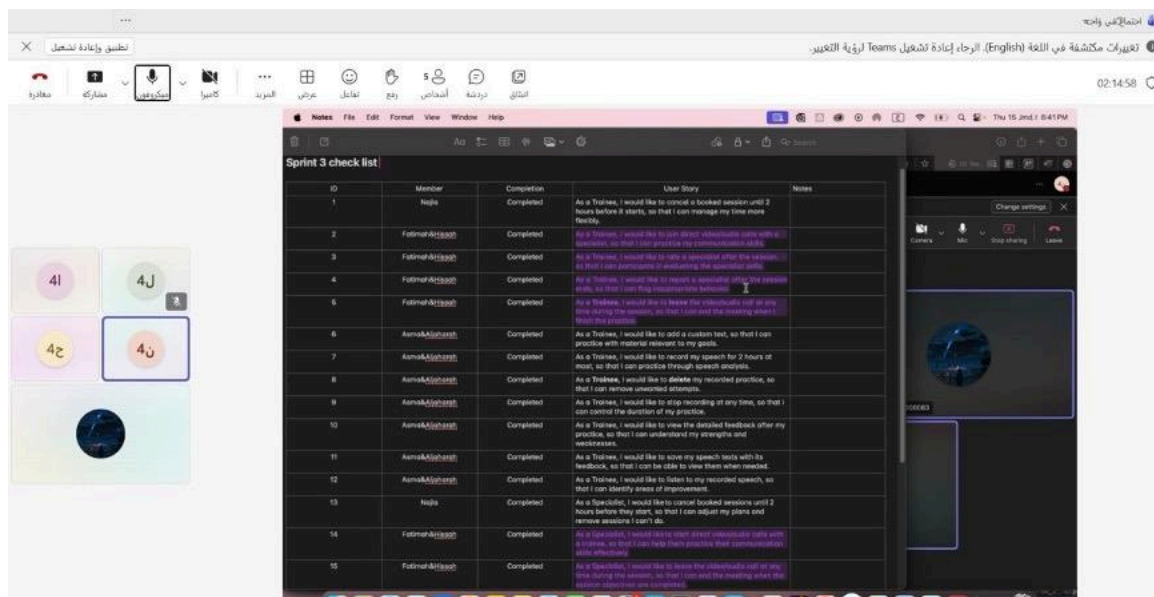


Figure [50]: checklist

